All results, first experiment to last

One result per row. Best framework = untrained tuned rules; Head A = trained speech path; Head B = trained picker; speech text = up to 600 transcript words for the answer model. dev = used to choose settings, test = locked and used once. Confirmed = the 95% interval excludes zero (green gain, red loss).

No.	Stage	Experiment	System (plain name)	Speech path (Path A)	Picture search (Path B)	Picker of the final pictures	Pictures shown	Text given to the answer model	Answer model	Trained parts used	Benchmark	Video type	Question set	Questions	Videos	What is measured	Result (%)	Correct / total	vs starting framework: difference (points)	vs starting framework: 95% interval	vs starting framework: confirmed?	vs best framework: difference (points)	vs best framework: 95% interval	vs best framework: confirmed?	What this row means
1	1. Version 1	v1 benchmark	v1: transcript only	BM25 windows (v1)	v1 candidate moments	no pictures	up to 8	BM25 excerpt, 120 words	Qwen3-VL-2B	none (controller rule hand-set)	LongVideoBench	everyday videos (8 s - 60 min)	validation (also used to build v1)	440	252	multiple-choice accuracy	32.5								v1: transcript only answered 32.5% of 440 LongVideoBench questions correctly. Answered from the transcript alone.
2	1. Version 1	v1 benchmark	v1: evenly spaced pictures	BM25 windows (v1)	v1 candidate moments	pictures spread evenly over the video	up to 8	BM25 excerpt, 120 words	Qwen3-VL-2B	none (controller rule hand-set)	LongVideoBench	everyday videos (8 s - 60 min)	validation (also used to build v1)	440	252	multiple-choice accuracy	39.3								v1: evenly spaced pictures answered 39.3% of 440 LongVideoBench questions correctly.
3	1. Version 1	v1 research idea	v1 controller trained to look when important speech is damaged	BM25 windows (v1)	v1 candidate moments	trained controller (tiny network; Qwen3-0.6B + LoRA)	up to 8	damaged transcript	Qwen3-VL-2B	v1 controller	LongVideoBench	everyday videos	pilot train	26	12	share of cases where damaging the important speech changed the answer	13.0								Only 13% (train) / 4% (dev, 14 questions) of damage cases mattered; the controller could not learn when to look. Idea not supported.
4	1. Version 1	v1 research check	Questions where losing one speech piece breaks the answer	-	-	-	-	clean vs damaged transcript	Qwen3-VL-2B	none	LongVideoBench		sample	50		share of questions where the answer depends on one local piece of speech	4.0	2 / 50							Only 2 of 50 LongVideoBench questions fit the v1 research idea, so it could not be tested.
5	1. Version 1	v1 research check	Questions where losing one speech piece breaks the answer	-	-	-	-	clean vs damaged transcript	Qwen3-VL-2B	none	Video-MME		sample	50		share of questions where the answer depends on one local piece of speech	4.0	2 / 50							Only 2 of 50 Video-MME questions fit the v1 research idea, so it could not be tested.
6	1. Version 1	v1 research check	Questions where losing one speech piece breaks the answer	-	-	-	-	clean vs damaged transcript	Qwen3-VL-2B	none	EduVidQA		sample	43		share of questions where the answer depends on one local piece of speech	0.0	0 / 43							Only 0 of 43 EduVidQA questions fit the v1 research idea, so it could not be tested.
7	1. Version 1	v1 research check	Questions where losing one speech piece breaks the answer	-	-	-	-	clean vs damaged transcript	Qwen3-VL-2B	none	TVQA		sample	50		share of questions where the answer depends on one local piece of speech	4.0	2 / 50							Only 2 of 50 TVQA questions fit the v1 research idea, so it could not be tested.
8	1. Version 1	v1 benchmark	v1: transcript-retrieved moments	BM25 windows (v1)	v1 candidate moments	start / middle / end of the best BM25 windows	up to 8	BM25 excerpt, 120 words	Qwen3-VL-2B	none (controller rule hand-set)	LongVideoBench	everyday videos (8 s - 60 min)	validation (also used to build v1)	440	252	multiple-choice accuracy	36.4								v1: transcript-retrieved moments answered 36.4% of 440 LongVideoBench questions correctly.
9	1. Version 1	v1 benchmark	v1: MobileCLIP top pictures	BM25 windows (v1)	v1 candidate moments	scout scores all candidates, take the most question-like	up to 8	BM25 excerpt, 120 words	Qwen3-VL-2B	none (controller rule hand-set)	LongVideoBench	everyday videos (8 s - 60 min)	validation (also used to build v1)	440	252	multiple-choice accuracy	41.6								v1: MobileCLIP top pictures answered 41.6% of 440 LongVideoBench questions correctly. Best v1 strategy; +9.1 over transcript only (confirmed).
10	1. Version 1	v1 benchmark	v1: cost-aware controller	BM25 windows (v1)	v1 candidate moments	controller: look / expand / stop by gain minus time cost	up to 8	BM25 excerpt, 120 words	Qwen3-VL-2B	none (controller rule hand-set)	LongVideoBench	everyday videos (8 s - 60 min)	validation (also used to build v1)	440	252	multiple-choice accuracy	37.5								v1: cost-aware controller answered 37.5% of 440 LongVideoBench questions correctly. The v1 research design; 4.1 below MobileCLIP (confirmed).
11	1. Version 1	v1 open-ended	v1: transcript only	BM25 windows (v1)	v1 candidate moments	as above	up to 8	BM25 excerpt, 120 words	Qwen3-VL-2B	none	LongVideoBench	everyday videos (8 s - 60 min)	validation (also used to build v1)	440	252	open-ended correct (local Phi-4-mini judge, never checked against humans)	44.1								44.1% judged correct; the judge was not validated, so these numbers were not used for claims.
12	1. Version 1	v1 open-ended	v1: evenly spaced pictures	BM25 windows (v1)	v1 candidate moments	as above	up to 8	BM25 excerpt, 120 words	Qwen3-VL-2B	none	LongVideoBench	everyday videos (8 s - 60 min)	validation (also used to build v1)	440	252	open-ended correct (local Phi-4-mini judge, never checked against humans)	33.4								33.4% judged correct; the judge was not validated, so these numbers were not used for claims.
13	1. Version 1	v1 open-ended	v1: transcript-retrieved moments	BM25 windows (v1)	v1 candidate moments	as above	up to 8	BM25 excerpt, 120 words	Qwen3-VL-2B	none	LongVideoBench	everyday videos (8 s - 60 min)	validation (also used to build v1)	440	252	open-ended correct (local Phi-4-mini judge, never checked against humans)	32.0								32.0% judged correct; the judge was not validated, so these numbers were not used for claims.
14	1. Version 1	v1 open-ended	v1: MobileCLIP top pictures	BM25 windows (v1)	v1 candidate moments	as above	up to 8	BM25 excerpt, 120 words	Qwen3-VL-2B	none	LongVideoBench	everyday videos (8 s - 60 min)	validation (also used to build v1)	440	252	open-ended correct (local Phi-4-mini judge, never checked against humans)	32.3								32.3% judged correct; the judge was not validated, so these numbers were not used for claims.

15	1. Version 1	v1 open-ended	v1: cost-aware controller	BM25 windows (v1)	v1 candidate moments	as above	up to 8	BM25 excerpt, 120 words	Qwen3-VL-2B	none	LongVideoBench	everyday videos (8 s - 60 min)	validation (also used to build v1)	440	252	open-ended correct (local Phi-4-mini judge, never checked against humans)	29.8								29.8% judged correct; the judge was not validated, so these numbers were not used for claims.
16	2. Version 2	v2 main experiment	v2: transcript only	speech rule inside BM25 windows	scan inside BM25 windows	no pictures	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	MIT lecture questions (self-made)	lecture videos (MIT OpenCourseWare)	each lecture tested by models that never saw it	197	20	multiple-choice accuracy	15.2								v2: transcript only: 15.2% of 197 MIT lecture questions. Below chance because of an answer-position bias in the generated questions.
17	2. Version 2	v2 main experiment	v2: MobileCLIP top-4	speech rule inside BM25 windows	scan inside BM25 windows	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	MIT lecture questions (self-made)	lecture videos (MIT OpenCourseWare)	each lecture tested by models that never saw it	197	20	multiple-choice accuracy	34.5								v2: MobileCLIP top-4: 34.5% of 197 MIT lecture questions.
18	2. Version 2	v2 main experiment	v2: MobileCLIP + diversity	speech rule inside BM25 windows	scan inside BM25 windows	MMR rule on the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	MIT lecture questions (self-made)	lecture videos (MIT OpenCourseWare)	each lecture tested by models that never saw it	197	20	multiple-choice accuracy	36.0								v2: MobileCLIP + diversity: 36.0% of 197 MIT lecture questions.
19	2. Version 2	v2 main experiment	v2: learned Head B with screen text	speech rule inside BM25 windows	scan inside BM25 windows	v2 Head B (trained)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	v2 Head B	MIT lecture questions (self-made)	lecture videos (MIT OpenCourseWare)	each lecture tested by models that never saw it	197	20	multiple-choice accuracy	34.7								v2: learned Head B with screen text: 34.7% of 197 MIT lecture questions. No gain over simple rules.
20	2. Version 2	v2 main experiment	v2: learned Head B without screen text	speech rule inside BM25 windows	scan inside BM25 windows	v2 Head B (trained)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	v2 Head B	MIT lecture questions (self-made)	lecture videos (MIT OpenCourseWare)	each lecture tested by models that never saw it	197	20	multiple-choice accuracy	32.8								v2: learned Head B without screen text: 32.8% of 197 MIT lecture questions.
21	2. Version 2	v2 main experiment	v2: learned Head A frame credit	speech rule inside BM25 windows	scan inside BM25 windows	v2 Head A (trained)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	v2 Head A	MIT lecture questions (self-made)	lecture videos (MIT OpenCourseWare)	each lecture tested by models that never saw it	197	20	multiple-choice accuracy	30.5								v2: learned Head A frame credit: 30.5% of 197 MIT lecture questions. Confirmed worse than the zero-shot rule.
22	2. Version 2	v2 main experiment	v2: zero-shot transcript relevance	speech rule inside BM25 windows	scan inside BM25 windows	top pictures by speech score (untrained)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	MIT lecture questions (self-made)	lecture videos (MIT OpenCourseWare)	each lecture tested by models that never saw it	197	20	multiple-choice accuracy	38.6								v2: zero-shot transcript relevance: 38.6% of 197 MIT lecture questions. Best v2 picker.
23	2. Version 2	v2 main experiment	v2: learned Head A text credit	speech rule inside BM25 windows	scan inside BM25 windows	v2 Head A text credit (trained)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	v2 Head A	MIT lecture questions (self-made)	lecture videos (MIT OpenCourseWare)	each lecture tested by models that never saw it	197	20	multiple-choice accuracy	37.1								v2: learned Head A text credit: 37.1% of 197 MIT lecture questions.
24	2. Version 2	v2 pilot	v2 learned pickers (pilot)	speech rule	scan inside windows	learned pickers	4	BM25 120 words	Qwen3-VL-2B	v2 heads	MIT lecture questions (self-made)	lectures	pilot	132	5	accuracy vs MobileCLIP top-4									Every learned picker was within noise of MobileCLIP top-4 (not supported).
25	2. Version 2	v2 follow-up	Learned 4th-picture completion	speech rule	scan inside windows	top-3 + learned 4th picture	4	BM25 120 words	Qwen3-VL-2B	completion scorer	MIT lecture questions (self-made)	lecture videos (MIT OpenCourseWare)	each lecture tested by models that never saw it	197	20	accuracy difference vs the rule's own 4th picture	-1.0								-1.0 points (interval -2.5 to 0.0): refuted.
26	2. Version 2	v2 follow-up	Picture scan spread evenly over windows	speech rule	balanced scan inside windows	top pictures by match to the question	4	BM25 120 words	Qwen3-VL-2B	none	MIT lecture questions (self-made)	lecture videos (MIT OpenCourseWare)	each lecture tested by models that never saw it	197	20	multiple-choice accuracy	39.1								39.1% vs 34.5% with the original scan (+4.6, interval -0.5 to +10.4): not confirmed.
27	2. Version 2	v2 follow-up	Option-rotation scoring, MobileCLIP top-4	speech rule	scan inside windows	top pictures by match	4	BM25 120 words	Qwen3-VL-2B	none	MIT lecture questions (self-made)	lecture videos (MIT OpenCourseWare)	each lecture tested by models that never saw it	197	20	accuracy averaged over the 4 option orders	49.7								49.7% averaged over option rotations vs 34.3% for transcript only: fixes the answer-position bias.
28	2. Version 2	v2 transfer check	v2 scan on LongVideoBench: original	speech rule	scan inside windows	MobileCLIP top-4	4	BM25 120 words	Qwen3-VL-2B	none	LongVideoBench	everyday videos	validation	440	252	multiple-choice accuracy	43.0								The MIT finding did not transfer to a public benchmark; v3 moved to public benchmarks.
29	2. Version 2	v2 transfer check	v2 scan on LongVideoBench: spread evenly	speech rule	scan inside windows	MobileCLIP top-4	4	BM25 120 words	Qwen3-VL-2B	none	LongVideoBench	everyday videos	validation	440	252	multiple-choice accuracy	42.5								The MIT finding did not transfer to a public benchmark; v3 moved to public benchmarks.
30	3. v3 benchmark check	v3 benchmark check	No pictures (transcript only)	-	-	No pictures (transcript only)	0	whole transcript at the question time	Qwen3-VL-2B	none	EduVidQA	lecture videos, open-ended	official train (checked only)	74	20	FactQA precision (share of supported facts, 0-1, not a percentage)	0.587								Pictures made the small model's lecture answers worse; EduVidQA was not used for picture claims.

31	3. v3 benchmark check	v3 benchmark check	4 evenly spaced pictures	-	-	4 evenly spaced pictures	4	whole transcript at the question time	Qwen3-VL-2B	none	EduVidQA	lecture videos, open-ended	official train (checked only)	74	20	FactQA precision (share of supported facts, 0-1, not a percentage)	0.493								Pictures made the small model's lecture answers worse; EduVidQA was not used for picture claims.
32	3. v3 benchmark check	v3 benchmark check	4 MobileCLIP-chosen pictures	-	-	4 MobileCLIP-chosen pictures	4	whole transcript at the question time	Qwen3-VL-2B	none	EduVidQA	lecture videos, open-ended	official train (checked only)	74	20	FactQA precision (share of supported facts, 0-1, not a percentage)	0.498								Pictures made the small model's lecture answers worse; EduVidQA was not used for picture claims.
33	3. v3 benchmark check	v3 benchmark check	Question only	-	-	Question only	0	whole transcript (2,000 words max)	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	97	97	multiple-choice accuracy	28.9								28.9%: pictures help on lectures (+16.5 for MobileCLIP pictures over transcript only, confirmed).
34	3. v3 benchmark check	v3 benchmark check	Transcript only	-	-	Transcript only	0	whole transcript (2,000 words max)	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	97	97	multiple-choice accuracy	40.2								40.2%: pictures help on lectures (+16.5 for MobileCLIP pictures over transcript only, confirmed).
35	3. v3 benchmark check	v3 benchmark check	4 evenly spaced pictures + transcript	-	-	4 evenly spaced pictures + transcript	4	whole transcript (2,000 words max)	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	97	97	multiple-choice accuracy	52.6								52.6%: pictures help on lectures (+16.5 for MobileCLIP pictures over transcript only, confirmed).
36	3. v3 benchmark check	v3 benchmark check	4 MobileCLIP pictures + transcript	-	-	4 MobileCLIP pictures + transcript	4	whole transcript (2,000 words max)	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	97	97	multiple-choice accuracy	56.7								56.7%: pictures help on lectures (+16.5 for MobileCLIP pictures over transcript only, confirmed).
37	3. v3 benchmark check	v3 benchmark check	Question only	-	-	Question only	0	Question only	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy	15.9								15.9%.
38	3. v3 benchmark check	v3 benchmark check	Transcript only (BM25 300 words)	-	-	Transcript only (BM25 300 words)	0	Transcript only (BM25 300 words)	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy	15.9								15.9%.
39	3. v3 benchmark check	v3 benchmark check	4 evenly spaced pictures	-	-	4 evenly spaced pictures	4	BM25 300 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy	17.1								17.1%.
40	3. v3 benchmark check	v3 benchmark check	4 MobileCLIP pictures (dense, whole video)	-	-	4 MobileCLIP pictures (dense, whole video)	4	BM25 300 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy	40.2								40.2%.
41	3. v3 benchmark check	v3 benchmark check	4 pictures from the human-marked answer seconds	-	-	4 pictures from the human-marked answer seconds	4	BM25 300 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy	63.4								63.4%. Ceiling for the 2B model with perfect pictures.
42	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: scan inside the windows, spread evenly; top pictures by match to the question	speech rule: MiniLM on lines inside the 4 BM25 windows	scan inside the windows, spread evenly	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	29.3	24 / 82	1.2	-6.7 to +10.7	no	-11.0	-18.6 to -2.9	yes (loss)	Tuning variant: scan inside the windows, spread evenly; top pictures by match to the question answered 29.3% of 82 CG-Bench dev questions correctly (2B answer model). +1.2 points vs the starting framework (not confirmed). -11.0 points vs the best framework (confirmed).
43	4. v3 tuning (dev)	evidence hit	Tuning variant: scan inside the windows, spread evenly; top pictures by match to the question	speech rule: MiniLM on lines inside the 4 BM25 windows	scan inside the windows, spread evenly	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	23.2	19 / 82							For 23.2% of these questions, Tuning variant: scan inside the windows, spread evenly; top pictures by match to the question showed a picture from the seconds humans marked as the answer.
44	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: scan inside the windows, spread evenly; top pictures by match to question + options	speech rule: MiniLM on lines inside the 4 BM25 windows	scan inside the windows, spread evenly	top pictures by match to question + options	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	30.5	25 / 82	2.4	-6.1 to +11.8	no	-9.8	-17.7 to -2.5	yes (loss)	Tuning variant: scan inside the windows, spread evenly; top pictures by match to question + options answered 30.5% of 82 CG-Bench dev questions correctly (2B answer model). +2.4 points vs the starting framework (not confirmed). -9.8 points vs the best framework (confirmed).
45	4. v3 tuning (dev)	evidence hit	Tuning variant: scan inside the windows, spread evenly; top pictures by match to question + options	speech rule: MiniLM on lines inside the 4 BM25 windows	scan inside the windows, spread evenly	top pictures by match to question + options	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	22.0	18 / 82							For 22.0% of these questions, Tuning variant: scan inside the windows, spread evenly; top pictures by match to question + options showed a picture from the seconds humans marked as the answer.
46	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: scan inside the windows, spread evenly; MMR rule on the question	speech rule: MiniLM on lines inside the 4 BM25 windows	scan inside the windows, spread evenly	MMR rule on the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	31.7	26 / 82	3.7	-3.3 to +12.5	no	-8.5	-16.7 to +0.0	no	Tuning variant: scan inside the windows, spread evenly; MMR rule on the question answered 31.7% of 82 CG-Bench dev questions correctly (2B answer model). +3.7 points vs the starting framework (not confirmed). -8.5 points vs the best framework (not confirmed).

47	4. v3 tuning (dev)	evidence hit	Tuning variant: scan inside the windows, spread evenly; MMR rule on the question	speech rule: MiniLM on lines inside the 4 BM25 windows	scan inside the windows, spread evenly	MMR rule on the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	24.4	20 / 82							For 24.4% of these questions, Tuning variant: scan inside the windows, spread evenly; MMR rule on the question showed a picture from the seconds humans marked as the answer.
48	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: scan inside the windows, spread evenly; MMR rule on question + options (best framework's picker)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan inside the windows, spread evenly	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	30.5	25 / 82	2.4	-5.1 to +12.7	no	-9.8	-18.3 to -1.3	yes (loss)	Tuning variant: scan inside the windows, spread evenly; MMR rule on question + options (best framework's picker) answered 30.5% of 82 CG-Bench dev questions correctly (2B answer model). +2.4 points vs the starting framework (not confirmed). -9.8 points vs the best framework (confirmed).
49	4. v3 tuning (dev)	evidence hit	Tuning variant: scan inside the windows, spread evenly; MMR rule on question + options (best framework's picker)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan inside the windows, spread evenly	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	24.4	20 / 82							For 24.4% of these questions, Tuning variant: scan inside the windows, spread evenly; MMR rule on question + options (best framework's picker) showed a picture from the seconds humans marked as the answer.
50	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: scan inside the windows, spread evenly; top pictures by speech score	speech rule: MiniLM on lines inside the 4 BM25 windows	scan inside the windows, spread evenly	top pictures by speech score	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	23.2	19 / 82	-4.9	-13.3 to +3.8	no	-17.1	-24.7 to -9.1	yes (loss)	Tuning variant: scan inside the windows, spread evenly; top pictures by speech score answered 23.2% of 82 CG-Bench dev questions correctly (2B answer model). -4.9 points vs the starting framework (not confirmed). -17.1 points vs the best framework (confirmed).
51	4. v3 tuning (dev)	evidence hit	Tuning variant: scan inside the windows, spread evenly; top pictures by speech score	speech rule: MiniLM on lines inside the 4 BM25 windows	scan inside the windows, spread evenly	top pictures by speech score	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	25.6	21 / 82							For 25.6% of these questions, Tuning variant: scan inside the windows, spread evenly; top pictures by speech score showed a picture from the seconds humans marked as the answer.
52	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: dense whole-video search (picture every 2 s); only speech-path pictures (best by question + options)	speech rule: MiniLM on lines inside the 4 BM25 windows	dense whole-video search (picture every 2 s)	only speech-path pictures (best by question + options)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	25.6	21 / 82	-2.4	-9.8 to +6.8	no	-14.6	-22.2 to -6.8	yes (loss)	Tuning variant: dense whole-video search (picture every 2 s); only speech-path pictures (best by question + options) answered 25.6% of 82 CG-Bench dev questions correctly (2B answer model). -2.4 points vs the starting framework (not confirmed). -14.6 points vs the best framework (confirmed).
53	4. v3 tuning (dev)	evidence hit	Tuning variant: dense whole-video search (picture every 2 s); only speech-path pictures (best by question + options)	speech rule: MiniLM on lines inside the 4 BM25 windows	dense whole-video search (picture every 2 s)	only speech-path pictures (best by question + options)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	24.4	20 / 82							For 24.4% of these questions, Tuning variant: dense whole-video search (picture every 2 s); only speech-path pictures (best by question + options) showed a picture from the seconds humans marked as the answer.
54	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: dense whole-video search (picture every 2 s); only picture-search pictures (best by question + options)	speech rule: MiniLM on lines inside the 4 BM25 windows	dense whole-video search (picture every 2 s)	only picture-search pictures (best by question + options)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	36.6	30 / 82	8.5	-1.3 to +19.1	no	-3.7	-12.0 to +3.7	no	Tuning variant: dense whole-video search (picture every 2 s); only picture-search pictures (best by question + options) answered 36.6% of 82 CG-Bench dev questions correctly (2B answer model). +8.5 points vs the starting framework (not confirmed). -3.7 points vs the best framework (not confirmed).
55	4. v3 tuning (dev)	evidence hit	Tuning variant: dense whole-video search (picture every 2 s); only picture-search pictures (best by question + options)	speech rule: MiniLM on lines inside the 4 BM25 windows	dense whole-video search (picture every 2 s)	only picture-search pictures (best by question + options)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	45.1	37 / 82							For 45.1% of these questions, Tuning variant: dense whole-video search (picture every 2 s); only picture-search pictures (best by question + options) showed a picture from the seconds humans marked as the answer.
56	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: dense whole-video search (picture every 2 s); top pictures by match to the question	speech rule: MiniLM on lines inside the 4 BM25 windows	dense whole-video search (picture every 2 s)	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	32.9	27 / 82	4.9	-4.8 to +14.3	no	-7.3	-15.2 to +0.0	no	Tuning variant: dense whole-video search (picture every 2 s); top pictures by match to the question answered 32.9% of 82 CG-Bench dev questions correctly (2B answer model). +4.9 points vs the starting framework (not confirmed). -7.3 points vs the best framework (not confirmed).
57	4. v3 tuning (dev)	evidence hit	Tuning variant: dense whole-video search (picture every 2 s); top pictures by match to the question	speech rule: MiniLM on lines inside the 4 BM25 windows	dense whole-video search (picture every 2 s)	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	46.3	38 / 82							For 46.3% of these questions, Tuning variant: dense whole-video search (picture every 2 s); top pictures by match to the question showed a picture from the seconds humans marked as the answer.

58	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: dense whole-video search (picture every 2 s); top pictures by match to question + options	speech rule: MiniLM on lines inside the 4 BM25 windows	dense whole-video search (picture every 2 s)	top pictures by match to question + options	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	37.8	31 / 82	9.8	-1.1 to +20.8	no	-2.4	-8.7 to +2.6	no	Tuning variant: dense whole-video search (picture every 2 s); top pictures by match to question + options answered 37.8% of 82 CG-Bench dev questions correctly (2B answer model). +9.8 points vs the starting framework (not confirmed). -2.4 points vs the best framework (not confirmed).
59	4. v3 tuning (dev)	evidence hit	Tuning variant: dense whole-video search (picture every 2 s); top pictures by match to question + options	speech rule: MiniLM on lines inside the 4 BM25 windows	dense whole-video search (picture every 2 s)	top pictures by match to question + options	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	50.0	41 / 82							For 50.0% of these questions, Tuning variant: dense whole-video search (picture every 2 s); top pictures by match to question + options showed a picture from the seconds humans marked as the answer.
60	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: dense whole-video search (picture every 2 s); MMR rule on the question	speech rule: MiniLM on lines inside the 4 BM25 windows	dense whole-video search (picture every 2 s)	MMR rule on the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	32.9	27 / 82	4.9	-2.7 to +14.5	no	-7.3	-13.5 to -2.4	yes (loss)	Tuning variant: dense whole-video search (picture every 2 s); MMR rule on the question answered 32.9% of 82 CG-Bench dev questions correctly (2B answer model). +4.9 points vs the starting framework (not confirmed). -7.3 points vs the best framework (confirmed).
61	4. v3 tuning (dev)	evidence hit	Tuning variant: dense whole-video search (picture every 2 s); MMR rule on the question	speech rule: MiniLM on lines inside the 4 BM25 windows	dense whole-video search (picture every 2 s)	MMR rule on the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	46.3	38 / 82							For 46.3% of these questions, Tuning variant: dense whole-video search (picture every 2 s); MMR rule on the question showed a picture from the seconds humans marked as the answer.
62	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: dense whole-video search (picture every 2 s); MMR rule on question + options (best framework's picker)	speech rule: MiniLM on lines inside the 4 BM25 windows	dense whole-video search (picture every 2 s)	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	34.1	28 / 82	6.1	-7.4 to +18.8	no	-6.1	-14.9 to +1.3	no	Tuning variant: dense whole-video search (picture every 2 s); MMR rule on question + options (best framework's picker) answered 34.1% of 82 CG-Bench dev questions correctly (2B answer model). +6.1 points vs the starting framework (not confirmed). -6.1 points vs the best framework (not confirmed).
63	4. v3 tuning (dev)	evidence hit	Tuning variant: dense whole-video search (picture every 2 s); MMR rule on question + options (best framework's picker)	speech rule: MiniLM on lines inside the 4 BM25 windows	dense whole-video search (picture every 2 s)	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	50.0	41 / 82							For 50.0% of these questions, Tuning variant: dense whole-video search (picture every 2 s); MMR rule on question + options (best framework's picker) showed a picture from the seconds humans marked as the answer.
64	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: dense whole-video search (picture every 2 s); top pictures by speech score	speech rule: MiniLM on lines inside the 4 BM25 windows	dense whole-video search (picture every 2 s)	top pictures by speech score	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	25.6	21 / 82	-2.4	-12.4 to +8.5	no	-14.6	-19.8 to -8.9	yes (loss)	Tuning variant: dense whole-video search (picture every 2 s); top pictures by speech score answered 25.6% of 82 CG-Bench dev questions correctly (2B answer model). -2.4 points vs the starting framework (not confirmed). -14.6 points vs the best framework (confirmed).
65	4. v3 tuning (dev)	evidence hit	Tuning variant: dense whole-video search (picture every 2 s); top pictures by speech score	speech rule: MiniLM on lines inside the 4 BM25 windows	dense whole-video search (picture every 2 s)	top pictures by speech score	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	28.0	23 / 82							For 28.0% of these questions, Tuning variant: dense whole-video search (picture every 2 s); top pictures by speech score showed a picture from the seconds humans marked as the answer.
66	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	only speech-path pictures (best by question + options)	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	30.5	25 / 82	2.4	-7.1 to +13.5	no	-9.8	-20.0 to +1.4	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options) answered 30.5% of 82 CG-Bench dev questions correctly (2B answer model). +2.4 points vs the starting framework (not confirmed). -9.8 points vs the best framework (not confirmed).
67	4. v3 tuning (dev)	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	only speech-path pictures (best by question + options)	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	24.4	20 / 82							For 24.4% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options) showed a picture from the seconds humans marked as the answer.

68	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	only speech-path pictures (best by question + options)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	25.6	21 / 82	-2.4	-9.8 to +6.8	no	-14.6	-22.2 to -6.8	yes (loss)	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options) answered 25.6% of 82 CG-Bench dev questions correctly (2B answer model). -2.4 points vs the starting framework (not confirmed). -14.6 points vs the best framework (confirmed).
69	4. v3 tuning (dev)	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	only speech-path pictures (best by question + options)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	24.4	20 / 82							For 24.4% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options) showed a picture from the seconds humans marked as the answer.
70	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only picture-search pictures (best by question + options)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	only picture-search pictures (best by question + options)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	34.1	28 / 82	6.1	-4.6 to +18.9	no	-6.1	-12.8 to +2.7	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only picture-search pictures (best by question + options) answered 34.1% of 82 CG-Bench dev questions correctly (2B answer model). +6.1 points vs the starting framework (not confirmed). -6.1 points vs the best framework (not confirmed).
71	4. v3 tuning (dev)	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only picture-search pictures (best by question + options)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	only picture-search pictures (best by question + options)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	43.9	36 / 82							For 43.9% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only picture-search pictures (best by question + options) showed a picture from the seconds humans marked as the answer.
72	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by match to the question	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	top pictures by match to the question	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	32.9	27 / 82	4.9	-2.4 to +13.2	no	-7.3	-12.6 to -1.3	yes (loss)	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by match to the question answered 32.9% of 82 CG-Bench dev questions correctly (2B answer model). +4.9 points vs the starting framework (not confirmed). -7.3 points vs the best framework (confirmed).
73	4. v3 tuning (dev)	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by match to the question	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	top pictures by match to the question	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	42.7	35 / 82							For 42.7% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by match to the question showed a picture from the seconds humans marked as the answer.
74	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by match to the question	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	29.3	24 / 82	1.2	-7.1 to +10.0	no	-11.0	-16.5 to -5.9	yes (loss)	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by match to the question answered 29.3% of 82 CG-Bench dev questions correctly (2B answer model). +1.2 points vs the starting framework (not confirmed). -11.0 points vs the best framework (confirmed).
75	4. v3 tuning (dev)	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by match to the question	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	42.7	35 / 82							For 42.7% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by match to the question showed a picture from the seconds humans marked as the answer.
76	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by match to question + options	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	top pictures by match to question + options	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	39.0	32 / 82	11.0	+0.0 to +22.4	no	-1.2	-6.1 to +4.2	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by match to question + options answered 39.0% of 82 CG-Bench dev questions correctly (2B answer model). +11.0 points vs the starting framework (not confirmed). -1.2 points vs the best framework (not confirmed).
77	4. v3 tuning (dev)	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by match to question + options	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	top pictures by match to question + options	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	46.3	38 / 82							For 46.3% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by match to question + options showed a picture from the seconds humans marked as the answer.

78	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by match to question + options	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	top pictures by match to question + options	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	34.1	28 / 82	6.1	-4.7 to +18.3	no	-6.1	-10.2 to -2.4	yes (loss)	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by match to question + options answered 34.1% of 82 CG-Bench dev questions correctly (2B answer model). +6.1 points vs the starting framework (not confirmed). -6.1 points vs the best framework (confirmed).
79	4. v3 tuning (dev)	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by match to question + options	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	top pictures by match to question + options	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	46.3	38 / 82							For 46.3% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by match to question + options showed a picture from the seconds humans marked as the answer.
80	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on the question	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on the question	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	39.0	32 / 82	11.0	+3.2 to +21.1	yes (gain)	-1.2	-5.1 to +2.7	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on the question answered 39.0% of 82 CG-Bench dev questions correctly (2B answer model). +11.0 points vs the starting framework (confirmed). -1.2 points vs the best framework (not confirmed).
81	4. v3 tuning (dev)	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on the question	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on the question	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	47.6	39 / 82							For 47.6% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on the question showed a picture from the seconds humans marked as the answer.
82	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on the question	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	32.9	27 / 82	4.9	-6.6 to +18.4	no	-7.3	-13.9 to +0.0	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on the question answered 32.9% of 82 CG-Bench dev questions correctly (2B answer model). +4.9 points vs the starting framework (not confirmed). -7.3 points vs the best framework (not confirmed).
83	4. v3 tuning (dev)	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on the question	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	47.6	39 / 82							For 47.6% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on the question showed a picture from the seconds humans marked as the answer.
84	4. v3 tuning (dev)	multiple-choice accuracy	Best framework	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	40.2	33 / 82	12.2	+4.3 to +21.3	yes (gain)				Best framework answered 40.2% of 82 CG-Bench dev questions correctly (2B answer model). +12.2 points vs the starting framework (confirmed).
85	4. v3 tuning (dev)	evidence hit	Best framework	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	50.0	41 / 82							For 50.0% of these questions, Best framework showed a picture from the seconds humans marked as the answer.
86	4. v3 tuning (dev)	open-ended correctness	Best framework	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	open-ended: the model writes its own answer (no options); AI judge marks it correct	19.5	16 / 82	6.1	-1.4 to +13.7	no				Best framework wrote a correct free answer for 19.5% of 82 CG-Bench dev questions.
87	4. v3 tuning (dev)	multiple-choice accuracy	Best framework + BM25 excerpt, 120 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	35.4	29 / 82	7.3	-2.3 to +18.5	no	-4.9	-7.7 to -1.4	yes (loss)	Best framework + BM25 excerpt, 120 words answered 35.4% of 82 CG-Bench dev questions correctly (2B answer model). +7.3 points vs the starting framework (not confirmed). -4.9 points vs the best framework (confirmed).
88	4. v3 tuning (dev)	evidence hit	Best framework + BM25 excerpt, 120 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	50.0	41 / 82							For 50.0% of these questions, Best framework + BM25 excerpt, 120 words showed a picture from the seconds humans marked as the answer.
89	4. v3 tuning (dev)	open-ended correctness	Best framework + BM25 excerpt, 120 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	open-ended: the model writes its own answer (no options); AI judge marks it correct	17.1	14 / 82	3.7	-2.2 to +10.3	no				Best framework + BM25 excerpt, 120 words wrote a correct free answer for 17.1% of 82 CG-Bench dev questions.

90	4. v3 tuning (dev)	multiple-choice accuracy	Best framework + BM25 excerpt, 300 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 300 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	34.1	28 / 82	6.1	-2.3 to +16.3	no	-6.1	-8.6 to -2.9	yes (loss)	Best framework + BM25 excerpt, 300 words answered 34.1% of 82 CG-Bench dev questions correctly (2B answer model). +6.1 points vs the starting framework (not confirmed). -6.1 points vs the best framework (confirmed).
91	4. v3 tuning (dev)	evidence hit	Best framework + BM25 excerpt, 300 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 300 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	50.0	41 / 82							For 50.0% of these questions, Best framework + BM25 excerpt, 300 words showed a picture from the seconds humans marked as the answer.
92	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by speech score	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	top pictures by speech score	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	25.6	21 / 82	-2.4	-12.4 to +8.5	no	-14.6	-19.8 to -8.9	yes (loss)	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by speech score answered 25.6% of 82 CG-Bench dev questions correctly (2B answer model). -2.4 points vs the starting framework (not confirmed). -14.6 points vs the best framework (confirmed).
93	4. v3 tuning (dev)	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by speech score	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	top pictures by speech score	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	29.3	24 / 82							For 29.3% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by speech score showed a picture from the seconds humans marked as the answer.
94	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on question + options (best framework's picker)	speech rule, only 3 pictures	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	37.8	31 / 82	9.8	+1.2 to +20.0	yes (gain)	-2.4	-5.4 to +0.0	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on question + options (best framework's picker) answered 37.8% of 82 CG-Bench dev questions correctly (2B answer model). +9.8 points vs the starting framework (confirmed). -2.4 points vs the best framework (not confirmed).
95	4. v3 tuning (dev)	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on question + options (best framework's picker)	speech rule, only 3 pictures	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	48.8	40 / 82							For 48.8% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on question + options (best framework's picker) showed a picture from the seconds humans marked as the answer.
96	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on question + options (best framework's picker)	speech rule, only 3 pictures	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	36.6	30 / 82	8.5	+1.0 to +18.8	yes (gain)	-3.7	-8.1 to +0.0	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on question + options (best framework's picker) answered 36.6% of 82 CG-Bench dev questions correctly (2B answer model). +8.5 points vs the starting framework (confirmed). -3.7 points vs the best framework (not confirmed).
97	4. v3 tuning (dev)	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on question + options (best framework's picker)	speech rule, only 3 pictures	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	48.8	40 / 82							For 48.8% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on question + options (best framework's picker) showed a picture from the seconds humans marked as the answer.
98	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid, 12 pictures instead of 6; MMR rule on question + options (best framework's picker)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid, 12 pictures instead of 6	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	36.6	30 / 82	8.5	-1.4 to +19.5	no	-3.7	-8.8 to +1.2	no	Tuning variant: hybrid, 12 pictures instead of 6; MMR rule on question + options (best framework's picker) answered 36.6% of 82 CG-Bench dev questions correctly (2B answer model). +8.5 points vs the starting framework (not confirmed). -3.7 points vs the best framework (not confirmed).
99	4. v3 tuning (dev)	evidence hit	Tuning variant: hybrid, 12 pictures instead of 6; MMR rule on question + options (best framework's picker)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid, 12 pictures instead of 6	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	48.8	40 / 82							For 48.8% of these questions, Tuning variant: hybrid, 12 pictures instead of 6; MMR rule on question + options (best framework's picker) showed a picture from the seconds humans marked as the answer.

100	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid, 12 pictures instead of 6; MMR rule on question + options (best framework's picker)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid, 12 pictures instead of 6	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	34.1	28 / 82	6.1	-2.4 to +15.8	no	-6.1	-12.2 to +0.0	no	Tuning variant: hybrid, 12 pictures instead of 6; MMR rule on question + options (best framework's picker) answered 34.1% of 82 CG-Bench dev questions correctly (2B answer model). +6.1 points vs the starting framework (not confirmed). -6.1 points vs the best framework (not confirmed).
101	4. v3 tuning (dev)	evidence hit	Tuning variant: hybrid, 12 pictures instead of 6; MMR rule on question + options (best framework's picker)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid, 12 pictures instead of 6	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	48.8	40 / 82							For 48.8% of these questions, Tuning variant: hybrid, 12 pictures instead of 6; MMR rule on question + options (best framework's picker) showed a picture from the seconds humans marked as the answer.
102	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: scan every 5 s inside the BM25 windows; only speech-path pictures (best by question + options)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	only speech-path pictures (best by question + options)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	25.6	21 / 82	-2.4	-9.8 to +6.8	no	-14.6	-22.2 to -6.8	yes (loss)	Tuning variant: scan every 5 s inside the BM25 windows; only speech-path pictures (best by question + options) answered 25.6% of 82 CG-Bench dev questions correctly (2B answer model). -2.4 points vs the starting framework (not confirmed). -14.6 points vs the best framework (confirmed).
103	4. v3 tuning (dev)	evidence hit	Tuning variant: scan every 5 s inside the BM25 windows; only speech-path pictures (best by question + options)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	only speech-path pictures (best by question + options)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	24.4	20 / 82							For 24.4% of these questions, Tuning variant: scan every 5 s inside the BM25 windows; only speech-path pictures (best by question + options) showed a picture from the seconds humans marked as the answer.
104	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: scan every 5 s inside the BM25 windows; only picture-search pictures (best by question + options)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	only picture-search pictures (best by question + options)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	30.5	25 / 82	2.4	-3.8 to +8.3	no	-9.8	-22.5 to +1.1	no	Tuning variant: scan every 5 s inside the BM25 windows; only picture-search pictures (best by question + options) answered 30.5% of 82 CG-Bench dev questions correctly (2B answer model). +2.4 points vs the starting framework (not confirmed). -9.8 points vs the best framework (not confirmed).
105	4. v3 tuning (dev)	evidence hit	Tuning variant: scan every 5 s inside the BM25 windows; only picture-search pictures (best by question + options)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	only picture-search pictures (best by question + options)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	18.3	15 / 82							For 18.3% of these questions, Tuning variant: scan every 5 s inside the BM25 windows; only picture-search pictures (best by question + options) showed a picture from the seconds humans marked as the answer.
106	4. v3 tuning (dev)	multiple-choice accuracy	Starting framework (v2 design)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	28.0	23 / 82				-12.2	-21.3 to -4.3	yes (loss)	Starting framework (v2 design) answered 28.0% of 82 CG-Bench dev questions correctly (2B answer model). -12.2 points vs the best framework (confirmed).
107	4. v3 tuning (dev)	evidence hit	Starting framework (v2 design)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	24.4	20 / 82							For 24.4% of these questions, Starting framework (v2 design) showed a picture from the seconds humans marked as the answer.
108	4. v3 tuning (dev)	open-ended correctness	Starting framework (v2 design)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	open-ended: the model writes its own answer (no options); AI judge marks it correct	13.4	11 / 82							Starting framework (v2 design) wrote a correct free answer for 13.4% of 82 CG-Bench dev questions.
109	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: scan every 5 s inside the BM25 windows; top pictures by match to question + options	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to question + options	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	26.8	22 / 82	-1.2	-8.4 to +5.9	no	-13.4	-21.1 to -7.1	yes (loss)	Tuning variant: scan every 5 s inside the BM25 windows; top pictures by match to question + options answered 26.8% of 82 CG-Bench dev questions correctly (2B answer model). -1.2 points vs the starting framework (not confirmed). -13.4 points vs the best framework (confirmed).
110	4. v3 tuning (dev)	evidence hit	Tuning variant: scan every 5 s inside the BM25 windows; top pictures by match to question + options	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to question + options	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	26.8	22 / 82							For 26.8% of these questions, Tuning variant: scan every 5 s inside the BM25 windows; top pictures by match to question + options showed a picture from the seconds humans marked as the answer.

111	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: scan every 5 s inside the BM25 windows; MMR rule on the question	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	MMR rule on the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	26.8	22 / 82	-1.2	-11.0 to +11.3	no	-13.4	-22.2 to -3.0	yes (loss)	Tuning variant: scan every 5 s inside the BM25 windows; MMR rule on the question answered 26.8% of 82 CG-Bench dev questions correctly (2B answer model). -1.2 points vs the starting framework (not confirmed). -13.4 points vs the best framework (confirmed).
112	4. v3 tuning (dev)	evidence hit	Tuning variant: scan every 5 s inside the BM25 windows; MMR rule on the question	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	MMR rule on the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	25.6	21 / 82							For 25.6% of these questions, Tuning variant: scan every 5 s inside the BM25 windows; MMR rule on the question showed a picture from the seconds humans marked as the answer.
113	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: scan every 5 s inside the BM25 windows; MMR rule on question + options (best framework's picker)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	28.0	23 / 82	0.0	-8.3 to +10.5	no	-12.2	-20.5 to -3.8	yes (loss)	Tuning variant: scan every 5 s inside the BM25 windows; MMR rule on question + options (best framework's picker) answered 28.0% of 82 CG-Bench dev questions correctly (2B answer model). +0.0 points vs the starting framework (not confirmed). -12.2 points vs the best framework (confirmed).
114	4. v3 tuning (dev)	evidence hit	Tuning variant: scan every 5 s inside the BM25 windows; MMR rule on question + options (best framework's picker)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	28.0	23 / 82							For 28.0% of these questions, Tuning variant: scan every 5 s inside the BM25 windows; MMR rule on question + options (best framework's picker) showed a picture from the seconds humans marked as the answer.
115	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: scan every 5 s inside the BM25 windows; top pictures by speech score	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by speech score	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	23.2	19 / 82	-4.9	-13.3 to +3.8	no	-17.1	-24.7 to -9.1	yes (loss)	Tuning variant: scan every 5 s inside the BM25 windows; top pictures by speech score answered 23.2% of 82 CG-Bench dev questions correctly (2B answer model). -4.9 points vs the starting framework (not confirmed). -17.1 points vs the best framework (confirmed).
116	4. v3 tuning (dev)	evidence hit	Tuning variant: scan every 5 s inside the BM25 windows; top pictures by speech score	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by speech score	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	25.6	21 / 82							For 25.6% of these questions, Tuning variant: scan every 5 s inside the BM25 windows; top pictures by speech score showed a picture from the seconds humans marked as the answer.
117	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: whole video, evenly (48 pictures); top pictures by match to the question	speech rule: MiniLM on lines inside the 4 BM25 windows	whole video, evenly (48 pictures)	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	34.1	28 / 82	6.1	-1.3 to +15.4	no	-6.1	-12.9 to +0.0	no	Tuning variant: whole video, evenly (48 pictures); top pictures by match to the question answered 34.1% of 82 CG-Bench dev questions correctly (2B answer model). +6.1 points vs the starting framework (not confirmed). -6.1 points vs the best framework (not confirmed).
118	4. v3 tuning (dev)	evidence hit	Tuning variant: whole video, evenly (48 pictures); top pictures by match to the question	speech rule: MiniLM on lines inside the 4 BM25 windows	whole video, evenly (48 pictures)	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	39.0	32 / 82							For 39.0% of these questions, Tuning variant: whole video, evenly (48 pictures); top pictures by match to the question showed a picture from the seconds humans marked as the answer.
119	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: whole video, evenly (48 pictures); top pictures by match to question + options	speech rule: MiniLM on lines inside the 4 BM25 windows	whole video, evenly (48 pictures)	top pictures by match to question + options	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	30.5	25 / 82	2.4	-5.4 to +12.7	no	-9.8	-15.5 to -4.0	yes (loss)	Tuning variant: whole video, evenly (48 pictures); top pictures by match to question + options answered 30.5% of 82 CG-Bench dev questions correctly (2B answer model). +2.4 points vs the starting framework (not confirmed). -9.8 points vs the best framework (confirmed).
120	4. v3 tuning (dev)	evidence hit	Tuning variant: whole video, evenly (48 pictures); top pictures by match to question + options	speech rule: MiniLM on lines inside the 4 BM25 windows	whole video, evenly (48 pictures)	top pictures by match to question + options	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	39.0	32 / 82							For 39.0% of these questions, Tuning variant: whole video, evenly (48 pictures); top pictures by match to question + options showed a picture from the seconds humans marked as the answer.
121	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: whole video, evenly (48 pictures); MMR rule on the question	speech rule: MiniLM on lines inside the 4 BM25 windows	whole video, evenly (48 pictures)	MMR rule on the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	29.3	24 / 82	1.2	-3.8 to +6.5	no	-11.0	-16.0 to -6.9	yes (loss)	Tuning variant: whole video, evenly (48 pictures); MMR rule on the question answered 29.3% of 82 CG-Bench dev questions correctly (2B answer model). +1.2 points vs the starting framework (not confirmed). -11.0 points vs the best framework (confirmed).

122	4. v3 tuning (dev)	evidence hit	Tuning variant: whole video, evenly (48 pictures); MMR rule on the question	speech rule: MiniLM on lines inside the 4 BM25 windows	whole video, evenly (48 pictures)	MMR rule on the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	39.0	32 / 82							For 39.0% of these questions, Tuning variant: whole video, evenly (48 pictures); MMR rule on the question showed a picture from the seconds humans marked as the answer.
123	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: whole video, evenly (48 pictures); MMR rule on question + options (best framework's picker)	speech rule: MiniLM on lines inside the 4 BM25 windows	whole video, evenly (48 pictures)	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	31.7	26 / 82	3.7	-3.5 to +13.3	no	-8.5	-13.5 to -3.8	yes (loss)	Tuning variant: whole video, evenly (48 pictures); MMR rule on question + options (best framework's picker) answered 31.7% of 82 CG-Bench dev questions correctly (2B answer model). +3.7 points vs the starting framework (not confirmed). -8.5 points vs the best framework (confirmed).
124	4. v3 tuning (dev)	evidence hit	Tuning variant: whole video, evenly (48 pictures); MMR rule on question + options (best framework's picker)	speech rule: MiniLM on lines inside the 4 BM25 windows	whole video, evenly (48 pictures)	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	36.6	30 / 82							For 36.6% of these questions, Tuning variant: whole video, evenly (48 pictures); MMR rule on question + options (best framework's picker) showed a picture from the seconds humans marked as the answer.
125	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: whole video, evenly (48 pictures); top pictures by speech score	speech rule: MiniLM on lines inside the 4 BM25 windows	whole video, evenly (48 pictures)	top pictures by speech score	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	22.0	18 / 82	-6.1	-14.6 to +4.2	no	-18.3	-23.5 to -11.3	yes (loss)	Tuning variant: whole video, evenly (48 pictures); top pictures by speech score answered 22.0% of 82 CG-Bench dev questions correctly (2B answer model). -6.1 points vs the starting framework (not confirmed). -18.3 points vs the best framework (confirmed).
126	4. v3 tuning (dev)	evidence hit	Tuning variant: whole video, evenly (48 pictures); top pictures by speech score	speech rule: MiniLM on lines inside the 4 BM25 windows	whole video, evenly (48 pictures)	top pictures by speech score	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	28.0	23 / 82							For 28.0% of these questions, Tuning variant: whole video, evenly (48 pictures); top pictures by speech score showed a picture from the seconds humans marked as the answer.
127	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: scan inside the windows, spread evenly; top pictures by match to the question	speech rule: MiniLM on lines inside the 4 BM25 windows	scan inside the windows, spread evenly	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	55.1	54 / 98	-1.0	-7.1 to +5.1	no	-8.2	-17.3 to +1.0	no	Tuning variant: scan inside the windows, spread evenly; top pictures by match to the question answered 55.1% of 98 Video-MMMU Perception dev questions correctly (2B answer model). -1.0 points vs the starting framework (not confirmed). -8.2 points vs the
128	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: scan inside the windows, spread evenly; top pictures by match to question + options	speech rule: MiniLM on lines inside the 4 BM25 windows	scan inside the windows, spread evenly	top pictures by match to question + options	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	57.1	56 / 98	1.0	-4.1 to +6.1	no	-6.1	-14.3 to +2.0	no	Tuning variant: scan inside the windows, spread evenly; top pictures by match to question + options answered 57.1% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +1.0 points vs the starting framework (not confirmed). -6.1 points vs the best framework (not confirmed).
129	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: scan inside the windows, spread evenly; MMR rule on the question	speech rule: MiniLM on lines inside the 4 BM25 windows	scan inside the windows, spread evenly	MMR rule on the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	57.1	56 / 98	1.0	-6.1 to +8.2	no	-6.1	-15.3 to +2.0	no	Tuning variant: scan inside the windows, spread evenly; MMR rule on the question answered 57.1% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +1.0 points vs the starting framework (not confirmed). -6.1 points vs the best framework (not confirmed).
130	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: scan inside the windows, spread evenly; MMR rule on question + options (best framework's picker)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan inside the windows, spread evenly	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	59.2	58 / 98	3.1	-5.1 to +11.2	no	-4.1	-11.2 to +3.1	no	Tuning variant: scan inside the windows, spread evenly; MMR rule on question + options (best framework's picker) answered 59.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +3.1 points vs the starting framework (not confirmed). -4.1 points vs the best framework (not confirmed).
131	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: scan inside the windows, spread evenly; top pictures by speech score	speech rule: MiniLM on lines inside the 4 BM25 windows	scan inside the windows, spread evenly	top pictures by speech score	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	51.0	50 / 98	-5.1	-12.2 to +1.0	no	-12.2	-21.4 to -3.1	yes (loss)	Tuning variant: scan inside the windows, spread evenly; top pictures by speech score answered 51.0% of 98 Video-MMMU Perception dev questions correctly (2B answer model). -5.1 points vs the starting framework (not confirmed). -12.2 points vs the best framework (confirmed).

132	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: dense whole-video search (picture every 2 s); only speech-path pictures (best by question + options)	speech rule: MiniLM on lines inside the 4 BM25 windows	dense whole-video search (picture every 2 s)	only speech-path pictures (best by question + options)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	55.1	54 / 98	-1.0	-8.2 to +5.1	no	-8.2	-17.3 to +1.0	no	Tuning variant: dense whole-video search (picture every 2 s); only speech-path pictures (best by question + options) answered 55.1% of 98 Video-MMMU Perception dev questions correctly (2B answer model). -1.0 points vs the starting framework (not confirmed). -8.2 points vs the best framework (not confirmed).
133	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: dense whole-video search (picture every 2 s); only picture-search pictures (best by question + options)	speech rule: MiniLM on lines inside the 4 BM25 windows	dense whole-video search (picture every 2 s)	only picture-search pictures (best by question + options)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	48.0	47 / 98	-8.2	-17.3 to +1.0	no	-15.3	-24.5 to -6.1	yes (loss)	Tuning variant: dense whole-video search (picture every 2 s); only picture-search pictures (best by question + options) answered 48.0% of 98 Video-MMMU Perception dev questions correctly (2B answer model). -8.2 points vs the starting framework (not confirmed). -15.3 points vs the best framework (confirmed).
134	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: dense whole-video search (picture every 2 s); top pictures by match to the question	speech rule: MiniLM on lines inside the 4 BM25 windows	dense whole-video search (picture every 2 s)	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	49.0	48 / 98	-7.1	-15.3 to +1.0	no	-14.3	-24.5 to -4.1	yes (loss)	Tuning variant: dense whole-video search (picture every 2 s); top pictures by match to the question answered 49.0% of 98 Video-MMMU Perception dev questions correctly (2B answer model). -7.1 points vs the starting framework (not confirmed). -14.3 points vs the best framework (confirmed).
135	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: dense whole-video search (picture every 2 s); top pictures by match to question + options	speech rule: MiniLM on lines inside the 4 BM25 windows	dense whole-video search (picture every 2 s)	top pictures by match to question + options	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	52.0	51 / 98	-4.1	-12.2 to +3.1	no	-11.2	-20.4 to -2.0	yes (loss)	Tuning variant: dense whole-video search (picture every 2 s); top pictures by match to question + options answered 52.0% of 98 Video-MMMU Perception dev questions correctly (2B answer model). -4.1 points vs the starting framework (not confirmed). -11.2 points vs the best framework (confirmed).
136	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: dense whole-video search (picture every 2 s); MMR rule on the question	speech rule: MiniLM on lines inside the 4 BM25 windows	dense whole-video search (picture every 2 s)	MMR rule on the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	55.1	54 / 98	-1.0	-7.1 to +6.1	no	-8.2	-16.3 to +0.0	no	Tuning variant: dense whole-video search (picture every 2 s); MMR rule on the question answered 55.1% of 98 Video-MMMU Perception dev questions correctly (2B answer model). -1.0 points vs the starting framework (not confirmed). -8.2 points vs the best framework (not confirmed).
137	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: dense whole-video search (picture every 2 s); MMR rule on question + options (best framework's picker)	speech rule: MiniLM on lines inside the 4 BM25 windows	dense whole-video search (picture every 2 s)	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	54.1	53 / 98	-2.0	-10.2 to +6.1	no	-9.2	-17.3 to -2.0	yes (loss)	Tuning variant: dense whole-video search (picture every 2 s); MMR rule on question + options (best framework's picker) answered 54.1% of 98 Video-MMMU Perception dev questions correctly (2B answer model). -2.0 points vs the starting framework (not confirmed). -9.2 points vs the best framework (confirmed).
138	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: dense whole-video search (picture every 2 s); top pictures by speech score	speech rule: MiniLM on lines inside the 4 BM25 windows	dense whole-video search (picture every 2 s)	top pictures by speech score	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	50.0	49 / 98	-6.1	-13.3 to +1.0	no	-13.3	-22.4 to -4.1	yes (loss)	Tuning variant: dense whole-video search (picture every 2 s); top pictures by speech score answered 50.0% of 98 Video-MMMU Perception dev questions correctly (2B answer model). -6.1 points vs the starting framework (not confirmed). -13.3 points vs the best framework (confirmed).
139	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	only speech-path pictures (best by question + options)	4	none	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	61.2	60 / 98	5.1	-3.1 to +13.3	no	-2.0	-10.2 to +6.1	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options) answered 61.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +5.1 points vs the starting framework (not confirmed). -2.0 points vs the best framework (not confirmed).

140	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	only speech-path pictures (best by question + options)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	55.1	54 / 98	-1.0	-8.2 to +5.1	no	-8.2	-17.3 to +1.0	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options) answered 55.1% of 98 Video-MMMU Perception dev questions correctly (2B answer model). -1.0 points vs the starting framework (not confirmed). -8.2 points vs the best framework (not confirmed).
141	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only picture-search pictures (best by question + options)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	only picture-search pictures (best by question + options)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	52.0	51 / 98	-4.1	-12.2 to +4.1	no	-11.2	-20.4 to -2.0	yes (loss)	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only picture-search pictures (best by question + options) answered 52.0% of 98 Video-MMMU Perception dev questions correctly (2B answer model). -4.1 points vs the starting framework (not confirmed). -11.2 points vs the best framework (confirmed).
142	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by match to the question	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	top pictures by match to the question	4	none	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	58.2	57 / 98	2.0	-5.1 to +9.2	no	-5.1	-12.2 to +2.0	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by match to the question answered 58.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +2.0 points vs the starting framework (not confirmed). -5.1 points vs the best framework (not confirmed).
143	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by match to the question	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	55.1	54 / 98	-1.0	-8.2 to +5.1	no	-8.2	-16.3 to +0.0	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by match to the question answered 55.1% of 98 Video-MMMU Perception dev questions correctly (2B answer model). -1.0 points vs the starting framework (not confirmed). -8.2 points vs the best framework (not confirmed).
144	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by match to question + options	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	top pictures by match to question + options	4	none	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	63.3	62 / 98	7.1	-1.0 to +15.3	no	0.0	-8.2 to +7.1	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by match to question + options answered 63.3% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +7.1 points vs the starting framework (not confirmed). +0.0 points vs the best framework (not confirmed).
145	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by match to question + options	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	top pictures by match to question + options	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	55.1	54 / 98	-1.0	-7.1 to +5.1	no	-8.2	-16.3 to -1.0	yes (loss)	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by match to question + options answered 55.1% of 98 Video-MMMU Perception dev questions correctly (2B answer model). -1.0 points vs the starting framework (not confirmed). -8.2 points vs the best framework (confirmed).
146	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on the question	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on the question	4	none	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	65.3	64 / 98	9.2	+2.0 to +16.3	yes (gain)	2.0	-3.1 to +7.1	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on the question answered 65.3% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +9.2 points vs the starting framework (confirmed). +2.0 points vs the best framework (not confirmed).
147	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on the question	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	58.2	57 / 98	2.0	-4.1 to +8.2	no	-5.1	-12.2 to +2.0	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on the question answered 58.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +2.0 points vs the starting framework (not confirmed). -5.1 points vs the best framework (not confirmed).
148	4. v3 tuning (dev)	multiple-choice accuracy	Best framework	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	63.3	62 / 98	7.1	-1.0 to +15.3	no				Best framework answered 63.3% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +7.1 points vs the starting framework (not confirmed).

149	4. v3 tuning (dev)	multiple-choice accuracy	Best framework + BM25 excerpt, 120 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	58.2	57 / 98	2.0	-5.1 to +9.2	no	-5.1	-11.2 to +1.0	no	Best framework + BM25 excerpt, 120 words answered 58.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +2.0 points vs the starting framework (not confirmed). -5.1 points vs the best framework (not confirmed).
150	4. v3 tuning (dev)	multiple-choice accuracy	Best framework + BM25 excerpt, 300 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 300 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	59.2	58 / 98	3.1	-4.1 to +11.2	no	-4.1	-9.2 to +0.0	no	Best framework + BM25 excerpt, 300 words answered 59.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +3.1 points vs the starting framework (not confirmed). -4.1 points vs the best framework (not confirmed).
151	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by speech score	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	top pictures by speech score	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	51.0	50 / 98	-5.1	-12.2 to +1.0	no	-12.2	-21.4 to -3.1	yes (loss)	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; top pictures by speech score answered 51.0% of 98 Video-MMMU Perception dev questions correctly (2B answer model). -5.1 points vs the starting framework (not confirmed). -12.2 points vs the best framework (confirmed).
152	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on question + options (best framework's picker)	speech rule, only 3 pictures	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	61.2	60 / 98	5.1	-2.0 to +13.3	no	-2.0	-7.1 to +3.1	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on question + options (best framework's picker) answered 61.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +5.1 points vs the starting framework (not confirmed). -2.0 points vs the best framework (not confirmed).
153	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on question + options (best framework's picker)	speech rule, only 3 pictures	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	57.1	56 / 98	1.0	-5.1 to +7.1	no	-6.1	-13.3 to +0.0	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on question + options (best framework's picker) answered 57.1% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +1.0 points vs the starting framework (not confirmed). -6.1 points vs the best framework (not confirmed).
154	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid, 12 pictures instead of 6; MMR rule on question + options (best framework's picker)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid, 12 pictures instead of 6	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	59.2	58 / 98	3.1	-5.1 to +11.2	no	-4.1	-10.2 to +1.0	no	Tuning variant: hybrid, 12 pictures instead of 6; MMR rule on question + options (best framework's picker) answered 59.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +3.1 points vs the starting framework (not confirmed). -4.1 points vs the best framework (not confirmed).
155	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid, 12 pictures instead of 6; MMR rule on question + options (best framework's picker)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid, 12 pictures instead of 6	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	56.1	55 / 98	0.0	-7.1 to +8.2	no	-7.1	-14.3 to +0.0	no	Tuning variant: hybrid, 12 pictures instead of 6; MMR rule on question + options (best framework's picker) answered 56.1% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +0.0 points vs the starting framework (not confirmed). -7.1 points vs the best framework (not confirmed).
156	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: scan every 5 s inside the BM25 windows; only speech-path pictures (best by question + options)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	only speech-path pictures (best by question + options)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	55.1	54 / 98	-1.0	-8.2 to +5.1	no	-8.2	-17.3 to +1.0	no	Tuning variant: scan every 5 s inside the BM25 windows; only speech-path pictures (best by question + options) answered 55.1% of 98 Video-MMMU Perception dev questions correctly (2B answer model). -1.0 points vs the starting framework (not confirmed). -8.2 points vs the best framework (not confirmed).
157	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: scan every 5 s inside the BM25 windows; only picture-search pictures (best by question + options)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	only picture-search pictures (best by question + options)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	51.0	50 / 98	-5.1	-14.3 to +3.1	no	-12.2	-23.5 to -2.0	yes (loss)	Tuning variant: scan every 5 s inside the BM25 windows; only picture-search pictures (best by question + options) answered 51.0% of 98 Video-MMMU Perception dev questions correctly (2B answer model). -5.1 points vs the starting framework (not confirmed). -12.2 points vs the best framework (confirmed).

158	4. v3 tuning (dev)	multiple-choice accuracy	Starting framework (v2 design)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	56.1	55 / 98				-7.1	-15.3 to +1.0	no	Starting framework (v2 design) answered 56.1% of 98 Video-MMMU Perception dev questions correctly (2B answer model). -7.1 points vs the best framework (not confirmed).
159	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: scan every 5 s inside the BM25 windows; top pictures by match to question + options	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to question + options	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	54.1	53 / 98	-2.0	-7.1 to +2.0	no	-9.2	-17.3 to -2.0	yes (loss)	Tuning variant: scan every 5 s inside the BM25 windows; top pictures by match to question + options answered 54.1% of 98 Video-MMMU Perception dev questions correctly (2B answer model). -2.0 points vs the starting framework (not confirmed). -9.2 points vs the best framework (confirmed).
160	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: scan every 5 s inside the BM25 windows; MMR rule on the question	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	MMR rule on the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	56.1	55 / 98	0.0	-7.1 to +7.1	no	-7.1	-16.3 to +1.0	no	Tuning variant: scan every 5 s inside the BM25 windows; MMR rule on the question answered 56.1% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +0.0 points vs the starting framework (not confirmed). -7.1 points vs the best framework (not confirmed).
161	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: scan every 5 s inside the BM25 windows; MMR rule on question + options (best framework's picker)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	58.2	57 / 98	2.0	-5.1 to +9.2	no	-5.1	-13.3 to +2.0	no	Tuning variant: scan every 5 s inside the BM25 windows; MMR rule on question + options (best framework's picker) answered 58.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +2.0 points vs the starting framework (not confirmed). -5.1 points vs the best framework (not confirmed).
162	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: scan every 5 s inside the BM25 windows; top pictures by speech score	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by speech score	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	51.0	50 / 98	-5.1	-12.2 to +1.0	no	-12.2	-21.4 to -3.1	yes (loss)	Tuning variant: scan every 5 s inside the BM25 windows; top pictures by speech score answered 51.0% of 98 Video-MMMU Perception dev questions correctly (2B answer model). -5.1 points vs the starting framework (not confirmed). -12.2 points vs the best framework (confirmed).
163	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: whole video, evenly (48 pictures); top pictures by match to the question	speech rule: MiniLM on lines inside the 4 BM25 windows	whole video, evenly (48 pictures)	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	57.1	56 / 98	1.0	-6.1 to +8.2	no	-6.1	-15.3 to +3.1	no	Tuning variant: whole video, evenly (48 pictures); top pictures by match to the question answered 57.1% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +1.0 points vs the starting framework (not confirmed). -6.1 points vs the best framework (not confirmed).
164	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: whole video, evenly (48 pictures); top pictures by match to question + options	speech rule: MiniLM on lines inside the 4 BM25 windows	whole video, evenly (48 pictures)	top pictures by match to question + options	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	59.2	58 / 98	3.1	-4.1 to +10.2	no	-4.1	-13.3 to +5.1	no	Tuning variant: whole video, evenly (48 pictures); top pictures by match to question + options answered 59.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +3.1 points vs the starting framework (not confirmed). -4.1 points vs the best framework (not confirmed).
165	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: whole video, evenly (48 pictures); MMR rule on the question	speech rule: MiniLM on lines inside the 4 BM25 windows	whole video, evenly (48 pictures)	MMR rule on the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	58.2	57 / 98	2.0	-6.1 to +11.2	no	-5.1	-14.3 to +3.1	no	Tuning variant: whole video, evenly (48 pictures); MMR rule on the question answered 58.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +2.0 points vs the starting framework (not confirmed). -5.1 points vs the best framework (not confirmed).
166	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: whole video, evenly (48 pictures); MMR rule on question + options (best framework's picker)	speech rule: MiniLM on lines inside the 4 BM25 windows	whole video, evenly (48 pictures)	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	58.2	57 / 98	2.0	-6.1 to +10.2	no	-5.1	-13.3 to +3.1	no	Tuning variant: whole video, evenly (48 pictures); MMR rule on question + options (best framework's picker) answered 58.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +2.0 points vs the starting framework (not confirmed). -5.1 points vs the best framework (not confirmed).

167	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: whole video, evenly (48 pictures); top pictures by speech score	speech rule: MiniLM on lines inside the 4 BM25 windows	whole video, evenly (48 pictures)	top pictures by speech score	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	48.0	47 / 98	-8.2	-15.3 to -1.0	yes (loss)	-15.3	-25.5 to -6.1	yes (loss)	Tuning variant: whole video, evenly (48 pictures); top pictures by speech score answered 48.0% of 98 Video-MMMU Perception dev questions correctly (2B answer model). -8.2 points vs the starting framework (confirmed). -15.3 points vs the best framework (confirmed).
168	4. v3 tuning (dev)	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	only speech-path pictures (best by question + options)	4	none	Qwen3-VL-2B	none	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	29.6	29 / 98	-3.1	-9.2 to +3.1	no	-2.0	-5.1 to +0.0	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options) answered 29.6% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). -3.1 points vs the starting framework (not confirmed). -2.0 points vs the best framework (not confirmed).
169	4. v3 tuning (dev)	multiple-choice accuracy	Best framework	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	none	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	31.6	31 / 98	-1.0	-7.1 to +5.1	no				Best framework answered 31.6% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). -1.0 points vs the starting framework (not confirmed).
170	4. v3 tuning (dev)	multiple-choice accuracy	Starting framework (v2 design)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	32.7	32 / 98				1.0	-5.1 to +7.1	no	Starting framework (v2 design) answered 32.7% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). +1.0 points vs the best framework (not confirmed).
171	5. Head A	Head A method screen	Speech rule inside BM25 windows (best framework's Path A)	Speech rule inside BM25 windows (best framework's Path A)	-	-	6 moments	-	-	none	CG-Bench	long everyday videos (10-80 min)	dev	164	17	evidence hit: at least one of 6 moments inside the human-marked answer seconds	25.0								25.0% of 164 new CG-Bench questions. Trained on 877 CG-Bench questions (83 videos).
172	5. Head A	Head A method screen	MiniLM over the whole transcript (untrained)	MiniLM over the whole transcript (untrained)	-	-	6 moments	-	-	none	CG-Bench	long everyday videos (10-80 min)	dev	164	17	evidence hit: at least one of 6 moments inside the human-marked answer seconds	28.0								28.0% of 164 new CG-Bench questions. Trained on 877 CG-Bench questions (83 videos). Reference for Head A.
173	5. Head A	Head A method screen	Fine-tuned MiniLM	Fine-tuned MiniLM	-	-	6 moments	-	-	fine-tuned MiniLM	CG-Bench	long everyday videos (10-80 min)	dev	164	17	evidence hit: at least one of 6 moments inside the human-marked answer seconds	8.0								8.0% of 164 new CG-Bench questions. Trained on 877 CG-Bench questions (83 videos). About 8%: training destroyed the ranking.
174	5. Head A	Head A method screen	Larger rerankers, untrained (best: BGE-base / BGE-v2-m3)	Larger rerankers, untrained (best: BGE-base / BGE-v2-m3)	-	-	6 moments	-	-	none	CG-Bench	long everyday videos (10-80 min)	dev	164	17	evidence hit: at least one of 6 moments inside the human-marked answer seconds	31.1								31.1% of 164 new CG-Bench questions. Trained on 877 CG-Bench questions (83 videos). Model size barely helps.
175	5. Head A	Head A method screen	Light trained network, with EduVidQA weak data	Light trained network, with EduVidQA weak data	-	-	6 moments	-	-	Head A draft	CG-Bench	long everyday videos (10-80 min)	dev	164	17	evidence hit: at least one of 6 moments inside the human-marked answer seconds	32.7								32.7% of 164 new CG-Bench questions. Trained on 877 CG-Bench questions (83 videos).
176	5. Head A	Head A method screen	Light trained network, human data only	Light trained network, human data only	-	-	6 moments	-	-	Head A draft	CG-Bench	long everyday videos (10-80 min)	dev	164	17	evidence hit: at least one of 6 moments inside the human-marked answer seconds	32.1								32.1% of 164 new CG-Bench questions. Trained on 877 CG-Bench questions (83 videos).
177	5. Head A	Head A method screen	Light network + lines vote	Light network + lines vote	-	-	6 moments	-	-	Head A draft	CG-Bench	long everyday videos (10-80 min)	dev	164	17	evidence hit: at least one of 6 moments inside the human-marked answer seconds	35.2								35.2% of 164 new CG-Bench questions. Trained on 877 CG-Bench questions (83 videos).
178	5. Head A	Head A method screen	Light network + learned offsets (original idea)	Light network + learned offsets (original idea)	-	-	6 moments	-	-	Head A draft	CG-Bench	long everyday videos (10-80 min)	dev	164	17	evidence hit: at least one of 6 moments inside the human-marked answer seconds	31.7								31.7% of 164 new CG-Bench questions. Trained on 877 CG-Bench questions (83 videos). Offsets did not help.
179	5. Head A	Head A method screen	Head A (chosen): lines vote + BGE facts, 3 copies averaged	Head A (chosen): lines vote + BGE facts, 3 copies averaged	-	-	6 moments	-	-	Head A	CG-Bench	long everyday videos (10-80 min)	dev	164	17	evidence hit: at least one of 6 moments inside the human-marked answer seconds	36.6								36.6% of 164 new CG-Bench questions. Trained on 877 CG-Bench questions (83 videos). +8.5 over 28.0 (interval +4.1 to +13.1), confirmed.

180	5. Head A	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options)	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	only speech-path pictures (best by question + options)	4	none	Qwen3-VL-2B	Head A	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	34.1	28 / 82	6.1	-2.1 to +16.2	no	-6.1	-12.9 to +2.6	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options) answered 34.1% of 82 CG-Bench dev questions correctly (2B answer model). +6.1 points vs the starting framework (not confirmed). -6.1 points vs the best framework (not confirmed).
181	5. Head A	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options)	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	only speech-path pictures (best by question + options)	4	none	Qwen3-VL-2B	Head A	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	28.0	23 / 82							For 28.0% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options) showed a picture from the seconds humans marked as the answer.
182	5. Head A	multiple-choice accuracy	Best framework + Head A	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	Head A	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	39.0	32 / 82	11.0	+3.8 to +20.3	yes (gain)	-1.2	-4.8 to +2.9	no	Best framework + Head A answered 39.0% of 82 CG-Bench dev questions correctly (2B answer model). +11.0 points vs the starting framework (confirmed). -1.2 points vs the best framework (not confirmed).
183	5. Head A	evidence hit	Best framework + Head A	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	Head A	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	45.1	37 / 82							For 45.1% of these questions, Best framework + Head A showed a picture from the seconds humans marked as the answer.
184	5. Head A	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options)	MiniLM over the whole transcript (untrained)	hybrid: 3 inside windows + 3 dense whole video	only speech-path pictures (best by question + options)	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	29.3	24 / 82	1.2	-12.8 to +17.8	no	-11.0	-24.2 to +1.4	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options) answered 29.3% of 82 CG-Bench dev questions correctly (2B answer model). +1.2 points vs the starting framework (not confirmed). -11.0 points vs the best framework (not confirmed).
185	5. Head A	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options)	MiniLM over the whole transcript (untrained)	hybrid: 3 inside windows + 3 dense whole video	only speech-path pictures (best by question + options)	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	30.5	25 / 82							For 30.5% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options) showed a picture from the seconds
186	5. Head A	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on question + options (best framework's picker)	MiniLM over the whole transcript (untrained)	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	39.0	32 / 82	11.0	+2.0 to +21.1	yes (gain)	-1.2	-3.6 to +0.0	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on question + options (best framework's picker) answered 39.0% of 82 CG-Bench dev questions correctly (2B answer model). +11.0 points vs the starting framework (confirmed). -1.2 points vs the best framework (not confirmed).
187	5. Head A	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on question + options (best framework's picker)	MiniLM over the whole transcript (untrained)	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	51.2	42 / 82							For 51.2% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on question + options (best framework's picker) showed a picture from the seconds humans marked as the answer.
188	5. Head A	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options)	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	only speech-path pictures (best by question + options)	4	none	Qwen3-VL-2B	Head A	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	57.1	56 / 98	1.0	-6.1 to +9.2	no	-6.1	-14.3 to +2.0	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options) answered 57.1% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +1.0 points vs the starting framework (not confirmed). -6.1 points vs the best framework (not confirmed).
189	5. Head A	multiple-choice accuracy	Best framework + Head A	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	Head A	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	62.2	61 / 98	6.1	-2.0 to +14.3	no	-1.0	-7.1 to +5.1	no	Best framework + Head A answered 62.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +6.1 points vs the starting framework (not confirmed). -1.0 points vs the best framework (not confirmed).

190	5. Head A	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options)	MiniLM over the whole transcript (untrained)	hybrid: 3 inside windows + 3 dense whole video	only speech-path pictures (best by question + options)	4	none	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	60.2	59 / 98	4.1	-4.1 to +12.2	no	-3.1	-11.2 to +5.1	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options) answered 60.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +4.1 points vs the starting framework (not confirmed). -3.1 points vs the best framework (not confirmed).
191	5. Head A	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on question + options (best framework's picker)	MiniLM over the whole transcript (untrained)	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	62.2	61 / 98	6.1	-2.0 to +14.3	no	-1.0	-3.1 to +0.0	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; MMR rule on question + options (best framework's picker) answered 62.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +6.1 points vs the starting framework (not confirmed). -1.0 points vs the best framework (not confirmed).
192	5. Head A	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options)	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	only speech-path pictures (best by question + options)	4	none	Qwen3-VL-2B	Head A	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	32.7	32 / 98	0.0	-6.1 to +6.1	no	1.0	-3.1 to +6.1	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; only speech-path pictures (best by question + options) answered 32.7% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). +0.0 points vs the starting framework (not confirmed). +1.0 points vs the best framework (not confirmed).
193	5. Head A	multiple-choice accuracy	Best framework + Head A	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	Head A	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	32.7	32 / 98	0.0	-6.1 to +6.1	no	1.0	-4.1 to +6.1	no	Best framework + Head A answered 32.7% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). +0.0 points vs the starting framework (not confirmed). +1.0 points vs the best framework (not confirmed).
194	6. Head B	Head B method screen (pooled dev)	Option-evidence set rule (untrained)	speech rule	hybrid	Option-evidence set rule (untrained)	4	none	Qwen3-VL-2B	none	CG-Bench + Video-MMMU (both tracks)	long videos and lectures	dev	278	113	accuracy difference to the MMR rule (points, pooled over 278 dev questions)	-1.4					-1.4		no	-1.4 points vs the MMR rule. Head B trained on 373 CG-Bench questions (37 videos).
195	6. Head B	Head B method screen (pooled dev)	Head B trained on human-marked seconds	speech rule	hybrid	Head B trained on human-marked seconds	4	none	Qwen3-VL-2B	Head B	CG-Bench + Video-MMMU (both tracks)	long videos and lectures	dev	278	113	accuracy difference to the MMR rule (points, pooled over 278 dev questions)	-1.4					-1.4		no	-1.4 points vs the MMR rule. Head B trained on 373 CG-Bench questions (37 videos).
196	6. Head B	Head B method screen (pooled dev)	Head B + option term (chosen Head B)	speech rule	hybrid	Head B + option term (chosen Head B)	4	none	Qwen3-VL-2B	Head B	CG-Bench + Video-MMMU (both tracks)	long videos and lectures	dev	278	113	accuracy difference to the MMR rule (points, pooled over 278 dev questions)	-0.4					-0.4		no	-0.4 points vs the MMR rule. Head B trained on 373 CG-Bench questions (37 videos).
197	6. Head B	Head B method screen (pooled dev)	Head B trained on answer-model feedback	speech rule	hybrid	Head B trained on answer-model feedback	4	none	Qwen3-VL-2B	Head B	CG-Bench + Video-MMMU (both tracks)	long videos and lectures	dev	278	113	accuracy difference to the MMR rule (points, pooled over 278 dev questions)	-3.6					-3.6		no	-3.6 points vs the MMR rule. Head B trained on 373 CG-Bench questions (37 videos). Worse on lectures.
198	6. Head B	Head B method screen (pooled dev)	Head B feedback + option term	speech rule	hybrid	Head B feedback + option term	4	none	Qwen3-VL-2B	Head B	CG-Bench + Video-MMMU (both tracks)	long videos and lectures	dev	278	113	accuracy difference to the MMR rule (points, pooled over 278 dev questions)	-4.0					-4.0		no	-4.0 points vs the MMR rule. Head B trained on 373 CG-Bench questions (37 videos).
199	6. Head B	Head B method screen (pooled dev)	Confidence choice over 4 picture sets	speech rule	hybrid	Confidence choice over 4 picture sets	4	none	Qwen3-VL-2B	none	CG-Bench + Video-MMMU (both tracks)	long videos and lectures	dev	278	113	accuracy difference to the MMR rule (points, pooled over 278 dev questions)	1.1					1.1		no	+1.1 points vs the MMR rule. Head B trained on 373 CG-Bench questions (37 videos). Interval -1.7 to +3.8.
200	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on human-marked seconds, no option term	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B trained on human-marked seconds, no option term	4	none	Qwen3-VL-2B	Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	35.4	29 / 82	7.3	-3.5 to +19.7	no	-4.9	-11.3 to +1.4	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on human-marked seconds, no option term answered 35.4% of 82 CG-Bench dev questions correctly (2B answer model). +7.3 points vs the starting framework (not confirmed). -4.9 points vs the best framework (not confirmed).

201	6. Head B	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on human-marked seconds, no option term	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B trained on human-marked seconds, no option term	4	none	Qwen3-VL-2B	Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	48.8	40 / 82							For 48.8% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on human-marked seconds, no option term showed a picture from the seconds humans marked as the answer.
202	6. Head B	multiple-choice accuracy	Best framework + Head B	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-2B	Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	39.0	32 / 82	11.0	+1.1 to +21.7	yes (gain)	-1.2	-7.2 to +5.2	no	Best framework + Head B answered 39.0% of 82 CG-Bench dev questions correctly (2B answer model). +11.0 points vs the starting framework (confirmed). -1.2 points vs the best framework (not confirmed).
203	6. Head B	evidence hit	Best framework + Head B	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-2B	Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	48.8	40 / 82							For 48.8% of these questions, Best framework + Head B showed a picture from the seconds humans marked as the answer.
204	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B without avoiding look-alikes	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B without avoiding look-alikes	4	none	Qwen3-VL-2B	Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	37.8	31 / 82	9.8	-1.0 to +21.1	no	-2.4	-7.6 to +2.9	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B without avoiding look-alikes answered 37.8% of 82 CG-Bench dev questions correctly (2B answer model). +9.8 points vs the starting framework (not confirmed). -2.4 points vs the best framework (not confirmed).
205	6. Head B	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B without avoiding look-alikes	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B without avoiding look-alikes	4	none	Qwen3-VL-2B	Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	48.8	40 / 82							For 48.8% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B without avoiding look-alikes showed a picture from the seconds humans marked as the answer.
206	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; option-evidence set rule (untrained)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	option-evidence set rule (untrained)	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	40.2	33 / 82	12.2	+1.1 to +24.3	yes (gain)	0.0	-6.0 to +7.7	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; option-evidence set rule (untrained) answered 40.2% of 82 CG-Bench dev questions correctly (2B answer model). +12.2 points vs the starting framework (confirmed). +0.0 points vs the best framework (not confirmed).
207	6. Head B	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; option-evidence set rule (untrained)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	option-evidence set rule (untrained)	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	47.6	39 / 82							For 47.6% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; option-evidence set rule (untrained) showed a picture from the seconds humans marked as the answer.
208	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on answer-model feedback	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B trained on answer-model feedback	4	none	Qwen3-VL-2B	Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	39.0	32 / 82	11.0	+0.0 to +22.5	no	-1.2	-7.0 to +5.3	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on answer-model feedback answered 39.0% of 82 CG-Bench dev questions correctly (2B answer model). +11.0 points vs the starting framework (not confirmed). -1.2 points vs the best framework (not confirmed).
209	6. Head B	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on answer-model feedback	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B trained on answer-model feedback	4	none	Qwen3-VL-2B	Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	51.2	42 / 82							For 51.2% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on answer-model feedback showed a picture from the seconds humans marked as the answer.
210	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B feedback version + option term	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B feedback version + option term	4	none	Qwen3-VL-2B	Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	39.0	32 / 82	11.0	+0.0 to +22.5	no	-1.2	-7.0 to +5.3	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B feedback version + option term answered 39.0% of 82 CG-Bench dev questions correctly (2B answer model). +11.0 points vs the starting framework (not confirmed). -1.2 points vs the best framework (not confirmed).
211	6. Head B	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B feedback version + option term	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B feedback version + option term	4	none	Qwen3-VL-2B	Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	51.2	42 / 82							For 51.2% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B feedback version + option term showed a picture from the seconds humans marked as the answer.

212	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on human-marked seconds, no option term	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B trained on human-marked seconds, no option term	4	none	Qwen3-VL-2B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	40.2	33 / 82	12.2	+3.4 to +21.1	yes (gain)	0.0	-4.6 to +5.4	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on human-marked seconds, no option term answered 40.2% of 82 CG-Bench dev questions correctly (2B answer model). +12.2 points vs the starting framework (confirmed). +0.0 points vs the best framework (not confirmed).
213	6. Head B	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on human-marked seconds, no option term	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B trained on human-marked seconds, no option term	4	none	Qwen3-VL-2B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	47.6	39 / 82							For 47.6% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on human-marked seconds, no option term showed a picture from the seconds humans marked as the answer.
214	6. Head B	multiple-choice accuracy	Best framework + Head A + Head B	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-2B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	40.2	33 / 82	12.2	+3.4 to +21.1	yes (gain)	0.0	-4.6 to +5.4	no	Best framework + Head A + Head B answered 40.2% of 82 CG-Bench dev questions correctly (2B answer model). +12.2 points vs the starting framework (confirmed). +0.0 points vs the best framework (not confirmed).
215	6. Head B	evidence hit	Best framework + Head A + Head B	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-2B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	47.6	39 / 82							For 47.6% of these questions, Best framework + Head A + Head B showed a picture from the seconds humans marked as the answer.
216	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B without avoiding look-alikes	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B without avoiding look-alikes	4	none	Qwen3-VL-2B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	40.2	33 / 82	12.2	+3.4 to +21.9	yes (gain)	0.0	-6.0 to +7.9	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B without avoiding look-alikes answered 40.2% of 82 CG-Bench dev questions correctly (2B answer model). +12.2 points vs the starting framework (confirmed). +0.0 points vs the best framework (not confirmed).
217	6. Head B	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B without avoiding look-alikes	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B without avoiding look-alikes	4	none	Qwen3-VL-2B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	47.6	39 / 82							For 47.6% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B without avoiding look-alikes showed a picture from the seconds humans marked as the answer.
218	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; option-evidence set rule (untrained)	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	option-evidence set rule (untrained)	4	none	Qwen3-VL-2B	Head A	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	42.7	35 / 82	14.6	+4.9 to +25.4	yes (gain)	2.4	-3.5 to +10.1	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; option-evidence set rule (untrained) answered 42.7% of 82 CG-Bench dev questions correctly (2B answer model). +14.6 points vs the starting framework (confirmed). +2.4 points vs the best framework (not confirmed).
219	6. Head B	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; option-evidence set rule (untrained)	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	option-evidence set rule (untrained)	4	none	Qwen3-VL-2B	Head A	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	48.8	40 / 82							For 48.8% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; option-evidence set rule (untrained) showed a picture from the seconds humans marked as the answer.
220	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on answer-model feedback	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B trained on answer-model feedback	4	none	Qwen3-VL-2B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	41.5	34 / 82	13.4	+2.4 to +26.0	yes (gain)	1.2	-4.8 to +9.9	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on answer-model feedback answered 41.5% of 82 CG-Bench dev questions correctly (2B answer model). +13.4 points vs the starting framework (confirmed). +1.2 points vs the best framework (not confirmed).
221	6. Head B	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on answer-model feedback	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B trained on answer-model feedback	4	none	Qwen3-VL-2B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	48.8	40 / 82							For 48.8% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on answer-model feedback showed a picture from the seconds humans marked as the answer.
222	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B feedback version + option term	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B feedback version + option term	4	none	Qwen3-VL-2B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	40.2	33 / 82	12.2	+1.1 to +25.4	yes (gain)	0.0	-7.1 to +9.6	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B feedback version + option term answered 40.2% of 82 CG-Bench dev questions correctly (2B answer model). +12.2 points vs the starting framework (confirmed). +0.0 points vs the best framework (not confirmed).

223	6. Head B	evidence hit	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B feedback version + option term	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B feedback version + option term	4	none	Qwen3-VL-2B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	47.6	39 / 82							For 47.6% of these questions, Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B feedback version + option term showed a picture from the seconds humans marked as the answer.
224	6. Head B	multiple-choice accuracy	Best framework + confidence choice (4 sets, most confident answer)	speech rule and Head A (both pools)	hybrid	confidence choice over 4 picture sets (most confident answer)	4 (per set)	none	Qwen3-VL-2B	Head A	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	42.7	35 / 82	14.6	+6.0 to +24.3	yes (gain)	2.4	+0.0 to +6.0	no	Best framework + confidence choice (4 sets, most confident answer) answered 42.7% of 82 CG-Bench dev questions correctly (2B answer model). +14.6 points vs the starting framework (confirmed). +2.4 points vs the best framework (not confirmed).
225	6. Head B	multiple-choice accuracy	Best framework + confidence choice (4 sets, summed probabilities)	speech rule and Head A (both pools)	hybrid	confidence choice over 4 picture sets (sum of option probabilities)	4 (per set)	none	Qwen3-VL-2B	Head A	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	41.5	34 / 82	13.4	+3.4 to +24.2	yes (gain)	1.2	-2.4 to +5.6	no	Best framework + confidence choice (4 sets, summed probabilities) answered 41.5% of 82 CG-Bench dev questions correctly (2B answer model). +13.4 points vs the starting framework (confirmed). +1.2 points vs the best framework (not confirmed).
226	6. Head B	multiple-choice accuracy	Best framework + confidence choice (6 sets, most confident answer)	speech rule and Head A (both pools)	hybrid	confidence choice over 6 picture sets (most confident answer)	4 (per set)	none	Qwen3-VL-2B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	42.7	35 / 82	14.6	+4.8 to +25.0	yes (gain)	2.4	-2.2 to +7.5	no	Best framework + confidence choice (6 sets, most confident answer) answered 42.7% of 82 CG-Bench dev questions correctly (2B answer model). +14.6 points vs the starting framework (confirmed). +2.4 points vs the best framework (not confirmed).
227	6. Head B	multiple-choice accuracy	Best framework + confidence choice (6 sets, summed probabilities)	speech rule and Head A (both pools)	hybrid	confidence choice over 6 picture sets (sum of option probabilities)	4 (per set)	none	Qwen3-VL-2B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	40.2	33 / 82	12.2	+0.0 to +24.1	no	0.0	-5.9 to +5.6	no	Best framework + confidence choice (6 sets, summed probabilities) answered 40.2% of 82 CG-Bench dev questions correctly (2B answer model). +12.2 points vs the starting framework (not confirmed). +0.0 points vs the best framework (not confirmed).
228	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on human-marked seconds, no option term	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B trained on human-marked seconds, no option term	4	none	Qwen3-VL-2B	Head B	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	62.2	61 / 98	6.1	-2.0 to +14.3	no	-1.0	-10.2 to +8.2	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on human-marked seconds, no option term answered 62.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +6.1 points vs the starting framework (not confirmed). -1.0 points vs the best framework (not confirmed).
229	6. Head B	multiple-choice accuracy	Best framework + Head B	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-2B	Head B	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	62.2	61 / 98	6.1	-2.0 to +14.3	no	-1.0	-10.2 to +8.2	no	Best framework + Head B answered 62.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +6.1 points vs the starting framework (not confirmed). -1.0 points vs the best framework (not confirmed).
230	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B without avoiding look-alikes	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B without avoiding look-alikes	4	none	Qwen3-VL-2B	Head B	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	60.2	59 / 98	4.1	-4.1 to +12.2	no	-3.1	-12.2 to +6.1	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B without avoiding look-alikes answered 60.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +4.1 points vs the starting framework (not confirmed). -3.1 points vs the best framework (not confirmed).
231	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; option-evidence set rule (untrained)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	option-evidence set rule (untrained)	4	none	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	63.3	62 / 98	7.1	-1.0 to +15.3	no	0.0	-8.2 to +7.1	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; option-evidence set rule (untrained) answered 63.3% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +7.1 points vs the starting framework (not confirmed). +0.0 points vs the best framework (not confirmed).
232	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on answer-model feedback	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B trained on answer-model feedback	4	none	Qwen3-VL-2B	Head B	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	53.1	52 / 98	-3.1	-11.2 to +5.1	no	-10.2	-20.4 to -1.0	yes (loss)	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on answer-model feedback answered 53.1% of 98 Video-MMMU Perception dev questions correctly (2B answer model). -3.1 points vs the starting framework (not confirmed). -10.2 points vs the best framework (confirmed).

233	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B feedback version + option term	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B feedback version + option term	4	none	Qwen3-VL-2B	Head B	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	52.0	51 / 98	-4.1	-13.3 to +4.1	no	-11.2	-21.4 to -2.0	yes (loss)	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B feedback version + option term answered 52.0% of 98 Video-MMMU Perception dev questions correctly (2B answer model). -4.1 points vs the starting framework (not confirmed). -11.2 points vs the best framework (confirmed).
234	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on human-marked seconds, no option term	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B trained on human-marked seconds, no option term	4	none	Qwen3-VL-2B	Head A, Head B	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	62.2	61 / 98	6.1	-1.0 to +14.3	no	-1.0	-10.2 to +8.2	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on human-marked seconds, no option term answered 62.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +6.1 points vs the starting framework (not confirmed). -1.0 points vs the best framework (not confirmed).
235	6. Head B	multiple-choice accuracy	Best framework + Head A + Head B	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-2B	Head A, Head B	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	62.2	61 / 98	6.1	-1.0 to +14.3	no	-1.0	-10.2 to +8.2	no	Best framework + Head A + Head B answered 62.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +6.1 points vs the starting framework (not confirmed). -1.0 points vs the best framework (not confirmed).
236	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B without avoiding look-alikes	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B without avoiding look-alikes	4	none	Qwen3-VL-2B	Head A, Head B	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	62.2	61 / 98	6.1	-1.0 to +14.3	no	-1.0	-10.2 to +8.2	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B without avoiding look-alikes answered 62.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +6.1 points vs the starting framework (not confirmed). -1.0 points vs the best framework (not confirmed).
237	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; option-evidence set rule (untrained)	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	option-evidence set rule (untrained)	4	none	Qwen3-VL-2B	Head A	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	60.2	59 / 98	4.1	-4.1 to +12.2	no	-3.1	-12.2 to +6.1	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; option-evidence set rule (untrained) answered 60.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +4.1 points vs the starting framework (not confirmed). -3.1 points vs the best framework (not confirmed).
238	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on answer-model feedback	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B trained on answer-model feedback	4	none	Qwen3-VL-2B	Head A, Head B	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	55.1	54 / 98	-1.0	-8.2 to +6.1	no	-8.2	-17.3 to +1.0	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on answer-model feedback answered 55.1% of 98 Video-MMMU Perception dev questions correctly (2B answer model). -1.0 points vs the starting framework (not confirmed). -8.2 points vs the best framework (not confirmed).
239	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B feedback version + option term	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B feedback version + option term	4	none	Qwen3-VL-2B	Head A, Head B	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	55.1	54 / 98	-1.0	-8.2 to +6.1	no	-8.2	-17.3 to +1.0	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B feedback version + option term answered 55.1% of 98 Video-MMMU Perception dev questions correctly (2B answer model). -1.0 points vs the starting framework (not confirmed). -8.2 points vs the best framework (not confirmed).
240	6. Head B	multiple-choice accuracy	Best framework + confidence choice (4 sets, most confident answer)	speech rule and Head A (both pools)	hybrid	confidence choice over 4 picture sets (most confident answer)	4 (per set)	none	Qwen3-VL-2B	Head A	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	66.3	65 / 98	10.2	+2.0 to +18.4	yes (gain)	3.1	-3.1 to +9.2	no	Best framework + confidence choice (4 sets, most confident answer) answered 66.3% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +10.2 points vs the starting framework (confirmed). +3.1 points vs the best framework (not confirmed).
241	6. Head B	multiple-choice accuracy	Best framework + confidence choice (4 sets, summed probabilities)	speech rule and Head A (both pools)	hybrid	confidence choice over 4 picture sets (sum of option probabilities)	4 (per set)	none	Qwen3-VL-2B	Head A	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	61.2	60 / 98	5.1	-2.0 to +12.2	no	-2.0	-7.1 to +3.1	no	Best framework + confidence choice (4 sets, summed probabilities) answered 61.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +5.1 points vs the starting framework (not confirmed). -2.0 points vs the best framework (not confirmed).

242	6. Head B	multiple-choice accuracy	Best framework + confidence choice (6 sets, most confident answer)	speech rule and Head A (both pools)	hybrid	confidence choice over 6 picture sets (most confident answer)	4 (per set)	none	Qwen3-VL-2B	Head A, Head B	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	63.3	62 / 98	7.1	+0.0 to +15.3	no	0.0	-7.2 to +7.1	no	Best framework + confidence choice (6 sets, most confident answer) answered 63.3% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +7.1 points vs the starting framework (not confirmed). +0.0 points vs the best framework (not confirmed).
243	6. Head B	multiple-choice accuracy	Best framework + confidence choice (6 sets, summed probabilities)	speech rule and Head A (both pools)	hybrid	confidence choice over 6 picture sets (sum of option probabilities)	4 (per set)	none	Qwen3-VL-2B	Head A, Head B	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	61.2	60 / 98	5.1	-1.0 to +12.2	no	-2.0	-9.2 to +5.1	no	Best framework + confidence choice (6 sets, summed probabilities) answered 61.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +5.1 points vs the starting framework (not confirmed). -2.0 points vs the best framework (not confirmed).
244	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on human-marked seconds, no option term	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B trained on human-marked seconds, no option term	4	none	Qwen3-VL-2B	Head B	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	32.7	32 / 98	0.0	-6.1 to +6.1	no	1.0	-3.1 to +6.1	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on human-marked seconds, no option term answered 32.7% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). +0.0 points vs the starting framework (not confirmed). +1.0 points vs the best framework (not confirmed).
245	6. Head B	multiple-choice accuracy	Best framework + Head B	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-2B	Head B	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	32.7	32 / 98	0.0	-6.1 to +6.1	no	1.0	-3.1 to +6.1	no	Best framework + Head B answered 32.7% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). +0.0 points vs the starting framework (not confirmed). +1.0 points vs the best framework (not confirmed).
246	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; option-evidence set rule (untrained)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	option-evidence set rule (untrained)	4	none	Qwen3-VL-2B	none	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	27.6	27 / 98	-5.1	-10.2 to +0.0	no	-4.1	-9.2 to +0.0	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; option-evidence set rule (untrained) answered 27.6% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). -5.1 points vs the starting framework (not confirmed). -4.1 points vs the best framework (not confirmed).
247	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on answer-model feedback	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B trained on answer-model feedback	4	none	Qwen3-VL-2B	Head B	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	32.7	32 / 98	0.0	-5.1 to +6.1	no	1.0	-3.1 to +5.1	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on answer-model feedback answered 32.7% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). +0.0 points vs the starting framework (not confirmed). +1.0 points vs the best framework (not confirmed).
248	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B feedback version + option term	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B feedback version + option term	4	none	Qwen3-VL-2B	Head B	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	32.7	32 / 98	0.0	-5.1 to +6.1	no	1.0	-3.1 to +5.1	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B feedback version + option term answered 32.7% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). +0.0 points vs the starting framework (not confirmed). +1.0 points vs the best framework (not confirmed).
249	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on human-marked seconds, no option term	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B trained on human-marked seconds, no option term	4	none	Qwen3-VL-2B	Head A, Head B	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	30.6	30 / 98	-2.0	-8.2 to +4.1	no	-1.0	-6.1 to +4.1	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on human-marked seconds, no option term answered 30.6% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). -2.0 points vs the starting framework (not confirmed). -1.0 points vs the best framework (not confirmed).
250	6. Head B	multiple-choice accuracy	Best framework + Head A + Head B	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-2B	Head A, Head B	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	30.6	30 / 98	-2.0	-8.2 to +4.1	no	-1.0	-6.1 to +4.1	no	Best framework + Head A + Head B answered 30.6% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). -2.0 points vs the starting framework (not confirmed). -1.0 points vs the best framework (not confirmed).

251	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; option-evidence set rule (untrained)	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	option-evidence set rule (untrained)	4	none	Qwen3-VL-2B	Head A	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	29.6	29 / 98	-3.1	-10.2 to +3.1	no	-2.0	-8.2 to +3.1	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; option-evidence set rule (untrained) answered 29.6% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). -3.1 points vs the starting framework (not confirmed). -2.0 points vs the best framework (not confirmed).
252	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on answer-model feedback	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B trained on answer-model feedback	4	none	Qwen3-VL-2B	Head A, Head B	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	34.7	34 / 98	2.0	-3.1 to +8.2	no	3.1	+0.0 to +7.1	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B trained on answer-model feedback answered 34.7% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). +2.0 points vs the starting framework (not confirmed). +3.1 points vs the best framework (not confirmed).
253	6. Head B	multiple-choice accuracy	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B feedback version + option term	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B feedback version + option term	4	none	Qwen3-VL-2B	Head A, Head B	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	34.7	34 / 98	2.0	-3.1 to +8.2	no	3.1	+0.0 to +7.1	no	Tuning variant: hybrid: 3 inside windows + 3 dense whole video; Head B feedback version + option term answered 34.7% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). +2.0 points vs the starting framework (not confirmed). +3.1 points vs the best framework (not confirmed).
254	6. Head B	multiple-choice accuracy	Best framework + confidence choice (4 sets, most confident answer)	speech rule and Head A (both pools)	hybrid	confidence choice over 4 picture sets (most confident answer)	4 (per set)	none	Qwen3-VL-2B	Head A	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	29.6	29 / 98	-3.1	-8.2 to +2.0	no	-2.0	-6.1 to +2.0	no	Best framework + confidence choice (4 sets, most confident answer) answered 29.6% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). -3.1 points vs the starting framework (not confirmed). -2.0 points vs the best framework (not confirmed).
255	6. Head B	multiple-choice accuracy	Best framework + confidence choice (4 sets, summed probabilities)	speech rule and Head A (both pools)	hybrid	confidence choice over 4 picture sets (sum of option probabilities)	4 (per set)	none	Qwen3-VL-2B	Head A	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	31.6	31 / 98	-1.0	-6.1 to +4.1	no	0.0	-3.1 to +3.1	no	Best framework + confidence choice (4 sets, summed probabilities) answered 31.6% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). -1.0 points vs the starting framework (not confirmed). +0.0 points vs the best framework (not confirmed).
256	7. Final locked test	multiple-choice accuracy	Standard rule: dense MobileCLIP top-4	none	dense whole-video search, top 4	top pictures by match to the question	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	41.8	23 / 55	7.3	+1.6 to +15.7	yes (gain)	-9.1	-15.5 to -2.0	yes (loss)	Standard rule: dense MobileCLIP top-4 answered 41.8% of 55 CG-Bench test questions correctly (2B answer model). +7.3 points vs the starting framework (confirmed). -9.1 points vs the best framework (confirmed).
257	7. Final locked test	evidence hit	Standard rule: dense MobileCLIP top-4	none	dense whole-video search, top 4	top pictures by match to the question	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	evidence hit: the shown pictures include the human-marked answer moment	58.2	32 / 55							For 58.2% of these questions, Standard rule: dense MobileCLIP top-4 showed a picture from the seconds humans marked as the answer.
258	7. Final locked test	open-ended correctness	Standard rule: dense MobileCLIP top-4	none	dense whole-video search, top 4	top pictures by match to the question	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	open-ended: the model writes its own answer (no options); AI judge marks it correct	25.5	14 / 55	7.3	-9.7 to +24.0	no				Standard rule: dense MobileCLIP top-4 wrote a correct free answer for 25.5% of 55 CG-Bench test questions.
259	7. Final locked test	multiple-choice accuracy	Best framework + Head B	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-2B	Head B	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	47.3	26 / 55	12.7	+4.9 to +22.0	yes (gain)	-3.6	-11.8 to +4.0	no	Best framework + Head B answered 47.3% of 55 CG-Bench test questions correctly (2B answer model). +12.7 points vs the starting framework (confirmed). -3.6 points vs the best framework (not confirmed).
260	7. Final locked test	evidence hit	Best framework + Head B	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-2B	Head B	CG-Bench	long everyday videos (10-80 min)	test	55	6	evidence hit: the shown pictures include the human-marked answer moment	50.9	28 / 55							For 50.9% of these questions, Best framework + Head B showed a picture from the seconds humans marked as the answer.
261	7. Final locked test	multiple-choice accuracy	Best framework	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	50.9	28 / 55	16.4	+7.0 to +26.0	yes (gain)				Best framework answered 50.9% of 55 CG-Bench test questions correctly (2B answer model). +16.4 points vs the starting framework (confirmed).

262	7. Final locked test	evidence hit	Best framework	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	evidence hit: the shown pictures include the human-marked answer moment	47.3	26 / 55							For 47.3% of these questions, Best framework showed a picture from the seconds humans marked as the answer.
263	7. Final locked test	open-ended correctness	Best framework	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	open-ended: the model writes its own answer (no options); AI judge marks it correct	23.6	13 / 55	5.5	-7.4 to +21.2	no				Best framework wrote a correct free answer for 23.6% of 55 CG-Bench test questions.
264	7. Final locked test	multiple-choice accuracy	No pictures (question and options only)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	no pictures	0	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	20.0	11 / 55	-14.5	-27.6 to -2.0	yes (loss)	-30.9	-37.5 to -22.6	yes (loss)	No pictures (question and options only) answered 20.0% of 55 CG-Bench test questions correctly (2B answer model). -14.5 points vs the starting framework (confirmed). -30.9 points vs the best framework (confirmed).
265	7. Final locked test	evidence hit	No pictures (question and options only)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	no pictures	0	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	evidence hit: the shown pictures include the human-marked answer moment	0.0	0 / 55							For 0.0% of these questions, No pictures (question and options only) showed a picture from the seconds humans marked as the answer.
266	7. Final locked test	multiple-choice accuracy	Best framework + Head A + Head B	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-2B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	34.5	19 / 55	0.0	-10.3 to +9.7	no	-16.4	-19.6 to -13.0	yes (loss)	Best framework + Head A + Head B answered 34.5% of 55 CG-Bench test questions correctly (2B answer model). +0.0 points vs the starting framework (not confirmed). -16.4 points vs the best framework (confirmed).
267	7. Final locked test	evidence hit	Best framework + Head A + Head B	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-2B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	test	55	6	evidence hit: the shown pictures include the human-marked answer moment	56.4	31 / 55							For 56.4% of these questions, Best framework + Head A + Head B showed a picture from the seconds humans marked as the answer.
268	7. Final locked test	multiple-choice accuracy	Best framework + Head A	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	Head A	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	41.8	23 / 55	7.3	-5.5 to +18.0	no	-9.1	-15.7 to -3.4	yes (loss)	Best framework + Head A answered 41.8% of 55 CG-Bench test questions correctly (2B answer model). +7.3 points vs the starting framework (not confirmed). -9.1 points vs the best framework (confirmed).
269	7. Final locked test	evidence hit	Best framework + Head A	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	Head A	CG-Bench	long everyday videos (10-80 min)	test	55	6	evidence hit: the shown pictures include the human-marked answer moment	49.1	27 / 55							For 49.1% of these questions, Best framework + Head A showed a picture from the seconds humans marked as the answer.
270	7. Final locked test	multiple-choice accuracy	Standard rule: 4 evenly spaced pictures	none	4 evenly spaced pictures	top pictures by match to the question	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	32.7	18 / 55	-1.8	-16.4 to +13.8	no	-18.2	-27.1 to -8.5	yes (loss)	Standard rule: 4 evenly spaced pictures answered 32.7% of 55 CG-Bench test questions correctly (2B answer model). -1.8 points vs the starting framework (not confirmed). -18.2 points vs the best framework (confirmed).
271	7. Final locked test	evidence hit	Standard rule: 4 evenly spaced pictures	none	4 evenly spaced pictures	top pictures by match to the question	4	none	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	evidence hit: the shown pictures include the human-marked answer moment	5.5	3 / 55							For 5.5% of these questions, Standard rule: 4 evenly spaced pictures showed a picture from the seconds humans marked as the answer.
272	7. Final locked test	multiple-choice accuracy	Starting framework (v2 design)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	34.5	19 / 55				-16.4	-26.0 to -7.0	yes (loss)	Starting framework (v2 design) answered 34.5% of 55 CG-Bench test questions correctly (2B answer model). -16.4 points vs the best framework (confirmed).
273	7. Final locked test	evidence hit	Starting framework (v2 design)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	evidence hit: the shown pictures include the human-marked answer moment	34.5	19 / 55							For 34.5% of these questions, Starting framework (v2 design) showed a picture from the seconds humans marked as the answer.
274	7. Final locked test	open-ended correctness	Starting framework (v2 design)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	open-ended: the model writes its own answer (no options); AI judge marks it correct	18.2	10 / 55							Starting framework (v2 design) wrote a correct free answer for 18.2% of 55 CG-Bench test questions.

275	7. Final locked test	multiple-choice accuracy	Best framework + confidence choice (4 sets, most confident answer)	speech rule and Head A (both pools)	hybrid	confidence choice over 4 picture sets (most confident answer)	4 (per set)	none	Qwen3-VL-2B	Head A	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	47.3	26 / 55	12.7	+0.0 to +24.1	no	-3.6	-7.7 to +0.0	no	Best framework + confidence choice (4 sets, most confident answer) answered 47.3% of 55 CG-Bench test questions correctly (2B answer model). +12.7 points vs the starting framework (not confirmed). -3.6 points vs the best framework (not confirmed).
276	7. Final locked test	multiple-choice accuracy	Best framework + confidence choice (4 sets, summed probabilities)	speech rule and Head A (both pools)	hybrid	confidence choice over 4 picture sets (sum of option probabilities)	4 (per set)	none	Qwen3-VL-2B	Head A	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	54.5	30 / 55	20.0	+8.9 to +33.3	yes (gain)	3.6	-3.4 to +9.8	no	Best framework + confidence choice (4 sets, summed probabilities) answered 54.5% of 55 CG-Bench test questions correctly (2B answer model). +20.0 points vs the starting framework (confirmed). +3.6 points vs the best framework (not confirmed).
277	7. Final locked test	multiple-choice accuracy	Standard rule: dense MobileCLIP top-4	none	dense whole-video search, top 4	top pictures by match to the question	4	none	Qwen3-VL-2B	none	Video-MMMU Comprehension	lectures, understanding questions	test	202	202	multiple-choice accuracy (answer letter correct)	31.7	64 / 202	1.5	-4.0 to +6.9	no	-0.5	-5.0 to +4.0	no	Standard rule: dense MobileCLIP top-4 answered 31.7% of 202 Video-MMMU Comprehension test questions correctly (2B answer model). +1.5 points vs the starting framework (not confirmed). -0.5 points vs the best framework (not confirmed).
278	7. Final locked test	multiple-choice accuracy	Best framework + Head B	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-2B	Head B	Video-MMMU Comprehension	lectures, understanding questions	test	202	202	multiple-choice accuracy (answer letter correct)	33.2	67 / 202	3.0	-2.5 to +8.4	no	1.0	-3.5 to +5.4	no	Best framework + Head B answered 33.2% of 202 Video-MMMU Comprehension test questions correctly (2B answer model). +3.0 points vs the starting framework (not confirmed). +1.0 points vs the best framework (not confirmed).
279	7. Final locked test	multiple-choice accuracy	Best framework	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	none	Video-MMMU Comprehension	lectures, understanding questions	test	202	202	multiple-choice accuracy (answer letter correct)	32.2	65 / 202	2.0	-3.5 to +6.9	no				Best framework answered 32.2% of 202 Video-MMMU Comprehension test questions correctly (2B answer model). +2.0 points vs the starting framework (not confirmed).
280	7. Final locked test	multiple-choice accuracy	No pictures (question and options only)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	no pictures	0	none	Qwen3-VL-2B	none	Video-MMMU Comprehension	lectures, understanding questions	test	202	202	multiple-choice accuracy (answer letter correct)	19.8	40 / 202	-10.4	-16.8 to -4.0	yes (loss)	-12.4	-18.8 to -5.9	yes (loss)	No pictures (question and options only) answered 19.8% of 202 Video-MMMU Comprehension test questions correctly (2B answer model). -10.4 points vs the starting framework (confirmed). -12.4 points vs the best framework (confirmed).
281	7. Final locked test	multiple-choice accuracy	Best framework + Head A + Head B	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-2B	Head A, Head B	Video-MMMU Comprehension	lectures, understanding questions	test	202	202	multiple-choice accuracy (answer letter correct)	32.7	66 / 202	2.5	-3.0 to +7.9	no	0.5	-4.0 to +5.0	no	Best framework + Head A + Head B answered 32.7% of 202 Video-MMMU Comprehension test questions correctly (2B answer model). +2.5 points vs the starting framework (not confirmed). +0.5 points vs the best framework (not confirmed).
282	7. Final locked test	multiple-choice accuracy	Best framework + Head A	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	Head A	Video-MMMU Comprehension	lectures, understanding questions	test	202	202	multiple-choice accuracy (answer letter correct)	33.7	68 / 202	3.5	-2.0 to +8.9	no	1.5	-2.5 to +5.4	no	Best framework + Head A answered 33.7% of 202 Video-MMMU Comprehension test questions correctly (2B answer model). +3.5 points vs the starting framework (not confirmed). +1.5 points vs the best framework (not confirmed).
283	7. Final locked test	multiple-choice accuracy	Standard rule: 4 evenly spaced pictures	none	4 evenly spaced pictures	top pictures by match to the question	4	none	Qwen3-VL-2B	none	Video-MMMU Comprehension	lectures, understanding questions	test	202	202	multiple-choice accuracy (answer letter correct)	33.2	67 / 202	3.0	-3.5 to +9.4	no	1.0	-4.5 to +6.4	no	Standard rule: 4 evenly spaced pictures answered 33.2% of 202 Video-MMMU Comprehension test questions correctly (2B answer model). +3.0 points vs the starting framework (not confirmed). +1.0 points vs the best framework (not confirmed).
284	7. Final locked test	multiple-choice accuracy	Starting framework (v2 design)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Comprehension	lectures, understanding questions	test	202	202	multiple-choice accuracy (answer letter correct)	30.2	61 / 202				-2.0	-6.9 to +3.5	no	Starting framework (v2 design) answered 30.2% of 202 Video-MMMU Comprehension test questions correctly (2B answer model). -2.0 points vs the best framework (not confirmed).
285	7. Final locked test	multiple-choice accuracy	Best framework + confidence choice (4 sets, most confident answer)	speech rule and Head A (both pools)	hybrid	confidence choice over 4 picture sets (most confident answer)	4 (per set)	none	Qwen3-VL-2B	Head A	Video-MMMU Comprehension	lectures, understanding questions	test	202	202	multiple-choice accuracy (answer letter correct)	33.2	67 / 202	3.0	-2.0 to +7.9	no	1.0	-3.0 to +5.0	no	Best framework + confidence choice (4 sets, most confident answer) answered 33.2% of 202 Video-MMMU Comprehension test questions correctly (2B answer model). +3.0 points vs the starting framework (not confirmed). +1.0 points vs the best framework (not confirmed).

286	7. Final locked test	multiple-choice accuracy	Best framework + confidence choice (4 sets, summed probabilities)	speech rule and Head A (both pools)	hybrid	confidence choice over 4 picture sets (sum of option probabilities)	4 (per set)	none	Qwen3-VL-2B	Head A	Video-MMMU Comprehension	lectures, understanding questions	test	202	202	multiple-choice accuracy (answer letter correct)	34.7	70 / 202	4.5	-0.5 to +9.9	no	2.5	-1.0 to +6.4	no	Best framework + confidence choice (4 sets, summed probabilities) answered 34.7% of 202 Video-MMMU Comprehension test questions correctly (2B answer model). +4.5 points vs the starting framework (not confirmed). +2.5 points vs the best framework (not confirmed).
287	7. Final locked test	multiple-choice accuracy	Standard rule: dense MobileCLIP top-4	none	dense whole-video search, top 4	top pictures by match to the question	4	none	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	test	202	202	multiple-choice accuracy (answer letter correct)	40.6	82 / 202	-6.4	-12.4 to -0.5	yes (loss)	-7.4	-12.9 to -2.5	yes (loss)	Standard rule: dense MobileCLIP top-4 answered 40.6% of 202 Video-MMMU Perception test questions correctly (2B answer model). -6.4 points vs the starting framework (confirmed). -7.4 points vs the best framework (confirmed).
288	7. Final locked test	multiple-choice accuracy	Best framework + Head B	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-2B	Head B	Video-MMMU Perception	lectures, screen-reading questions	test	202	202	multiple-choice accuracy (answer letter correct)	48.0	97 / 202	1.0	-4.0 to +5.9	no	0.0	-4.5 to +4.5	no	Best framework + Head B answered 48.0% of 202 Video-MMMU Perception test questions correctly (2B answer model). +1.0 points vs the starting framework (not confirmed). +0.0 points vs the best framework (not confirmed).
289	7. Final locked test	multiple-choice accuracy	Best framework	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	test	202	202	multiple-choice accuracy (answer letter correct)	48.0	97 / 202	1.0	-5.0 to +6.9	no				Best framework answered 48.0% of 202 Video-MMMU Perception test questions correctly (2B answer model). +1.0 points vs the starting framework (not confirmed).
290	7. Final locked test	multiple-choice accuracy	No pictures (question and options only)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	no pictures	0	none	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	test	202	202	multiple-choice accuracy (answer letter correct)	25.2	51 / 202	-21.8	-28.7 to -14.4	yes (loss)	-22.8	-30.2 to -14.9	yes (loss)	No pictures (question and options only) answered 25.2% of 202 Video-MMMU Perception test questions correctly (2B answer model). -21.8 points vs the starting framework (confirmed). -22.8 points vs the best framework (confirmed).
291	7. Final locked test	multiple-choice accuracy	Best framework + Head A + Head B	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-2B	Head A, Head B	Video-MMMU Perception	lectures, screen-reading questions	test	202	202	multiple-choice accuracy (answer letter correct)	52.0	105 / 202	5.0	+0.0 to +9.9	no	4.0	-0.5 to +8.4	no	Best framework + Head A + Head B answered 52.0% of 202 Video-MMMU Perception test questions correctly (2B answer model). +5.0 points vs the starting framework (not confirmed). +4.0 points vs the best framework (not confirmed).
292	7. Final locked test	multiple-choice accuracy	Best framework + Head A	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	Head A	Video-MMMU Perception	lectures, screen-reading questions	test	202	202	multiple-choice accuracy (answer letter correct)	45.0	91 / 202	-2.0	-8.4 to +4.0	no	-3.0	-7.4 to +1.5	no	Best framework + Head A answered 45.0% of 202 Video-MMMU Perception test questions correctly (2B answer model). -2.0 points vs the starting framework (not confirmed). -3.0 points vs the best framework (not confirmed).
293	7. Final locked test	multiple-choice accuracy	Standard rule: 4 evenly spaced pictures	none	4 evenly spaced pictures	top pictures by match to the question	4	none	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	test	202	202	multiple-choice accuracy (answer letter correct)	40.6	82 / 202	-6.4	-12.4 to +0.0	no	-7.4	-13.4 to -1.0	yes (loss)	Standard rule: 4 evenly spaced pictures answered 40.6% of 202 Video-MMMU Perception test questions correctly (2B answer model). -6.4 points vs the starting framework (not confirmed). -7.4 points vs the best framework (confirmed).
294	7. Final locked test	multiple-choice accuracy	Starting framework (v2 design)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	test	202	202	multiple-choice accuracy (answer letter correct)	47.0	95 / 202				-1.0	-6.9 to +5.0	no	Starting framework (v2 design) answered 47.0% of 202 Video-MMMU Perception test questions correctly (2B answer model). -1.0 points vs the best framework (not confirmed).
295	7. Final locked test	multiple-choice accuracy	Best framework + confidence choice (4 sets, most confident answer)	speech rule and Head A (both pools)	hybrid	confidence choice over 4 picture sets (most confident answer)	4 (per set)	none	Qwen3-VL-2B	Head A	Video-MMMU Perception	lectures, screen-reading questions	test	202	202	multiple-choice accuracy (answer letter correct)	50.0	101 / 202	3.0	-2.5 to +8.4	no	2.0	-2.0 to +5.9	no	Best framework + confidence choice (4 sets, most confident answer) answered 50.0% of 202 Video-MMMU Perception test questions correctly (2B answer model). +3.0 points vs the starting framework (not confirmed). +2.0 points vs the best framework (not confirmed).
296	7. Final locked test	multiple-choice accuracy	Best framework + confidence choice (4 sets, summed probabilities)	speech rule and Head A (both pools)	hybrid	confidence choice over 4 picture sets (sum of option probabilities)	4 (per set)	none	Qwen3-VL-2B	Head A	Video-MMMU Perception	lectures, screen-reading questions	test	202	202	multiple-choice accuracy (answer letter correct)	48.0	97 / 202	1.0	-4.5 to +5.9	no	0.0	-3.5 to +3.5	no	Best framework + confidence choice (4 sets, summed probabilities) answered 48.0% of 202 Video-MMMU Perception test questions correctly (2B answer model). +1.0 points vs the starting framework (not confirmed). +0.0 points vs the best framework (not confirmed).

297	8. Speech text	speech kept	v2 BM25 excerpt, 120 words	speech rule	hybrid	MMR rule on question + options	4	v2 BM25 excerpt, 120 words	-	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	speech kept: the given words include speech within 10 s of the human-marked answer moment	28.0								28.0% of questions: the text given to the answer model contains the speech near the answer.
298	8. Speech text	speech kept	No words	speech rule	hybrid	MMR rule on question + options	4	No words	-	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	speech kept: the given words include speech within 10 s of the human-marked answer moment	0.0								0.0% of questions: the text given to the answer model contains the speech near the answer.
299	8. Speech text	speech kept	Meaning-based search, 120 words	speech rule	hybrid	MMR rule on question + options	4	Meaning-based search, 120 words	-	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	speech kept: the given words include speech within 10 s of the human-marked answer moment	50.0								50.0% of questions: the text given to the answer model contains the speech near the answer.
300	8. Speech text	speech kept	Meaning-based search, 300 words	speech rule	hybrid	MMR rule on question + options	4	Meaning-based search, 300 words	-	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	speech kept: the given words include speech within 10 s of the human-marked answer moment	62.2								62.2% of questions: the text given to the answer model contains the speech near the answer.
301	8. Speech text	speech kept	Speech around the pictures, 300 words	speech rule	hybrid	MMR rule on question + options	4	Speech around the pictures, 300 words	-	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	speech kept: the given words include speech within 10 s of the human-marked answer moment	48.8								48.8% of questions: the text given to the answer model contains the speech near the answer.
302	8. Speech text	speech kept	Speech around pictures + search, 300 words	speech rule	hybrid	MMR rule on question + options	4	Speech around pictures + search, 300 words	-	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	speech kept: the given words include speech within 10 s of the human-marked answer moment	65.9								65.9% of questions: the text given to the answer model contains the speech near the answer.
303	8. Speech text	speech kept	Speech around pictures + search, 600 words (chosen)	speech rule	hybrid	MMR rule on question + options	4	Speech around pictures + search, 600 words (chosen)	-	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	speech kept: the given words include speech within 10 s of the human-marked answer moment	68.3								68.3% of questions: the text given to the answer model contains the speech near the answer.
304	8. Speech text	speech kept	Whole transcript, max 1,500 words	speech rule	hybrid	MMR rule on question + options	4	Whole transcript, max 1,500 words	-	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	speech kept: the given words include speech within 10 s of the human-marked answer moment	65.9								65.9% of questions: the text given to the answer model contains the speech near the answer.
305	8. Speech text	speech kept	Whole transcript, max 4,000 words	speech rule	hybrid	MMR rule on question + options	4	Whole transcript, max 4,000 words	-	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	speech kept: the given words include speech within 10 s of the human-marked answer moment	79.3								79.3% of questions: the text given to the answer model contains the speech near the answer.
306	8. Speech text	speech kept	Whole transcript, no cap	speech rule	hybrid	MMR rule on question + options	4	Whole transcript, no cap	-	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	speech kept: the given words include speech within 10 s of the human-marked answer moment	80.5								80.5% of questions: the text given to the answer model contains the speech near the answer.
307	8. Speech text	multiple-choice accuracy	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 300 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	36.6	30 / 82	8.5	+1.1 to +17.9	yes (gain)	-3.7	-6.5 to +0.0	no	Best framework + speech text answered 36.6% of 82 CG-Bench dev questions correctly (2B answer model). +8.5 points vs the starting framework (confirmed). -3.7 points vs the best framework (not confirmed).
308	8. Speech text	evidence hit	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 300 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	50.0	41 / 82							For 50.0% of these questions, Best framework + speech text showed a picture from the seconds humans marked as the answer.

309	8. Speech text	multiple-choice accuracy	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	37.8	31 / 82	9.8	+1.1 to +20.0	yes (gain)				Best framework + speech text answered 37.8% of 82 CG-Bench dev questions correctly (2B answer model). +9.8 points vs the starting framework (confirmed).
310	8. Speech text	evidence hit	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	50.0	41 / 82							For 50.0% of these questions, Best framework + speech text showed a picture from the seconds humans marked as the answer.
311	8. Speech text	open-ended correctness	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	open-ended: the model writes its own answer (no options); AI judge marks it correct	22.0	18 / 82	8.5	+3.3 to +14.8	yes (gain)				Best framework + speech text wrote a correct free answer for 22.0% of 82 CG-Bench dev questions.
312	8. Speech text	multiple-choice accuracy	Best framework + whole transcript, capped at 100000 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	whole transcript, capped at 100000 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	6	3	multiple-choice accuracy (answer letter correct)	50.0	3 / 6							Best framework + whole transcript, capped at 100000 words answered 50.0% of 6 CG-Bench dev questions correctly (2B answer model).
313	8. Speech text	evidence hit	Best framework + whole transcript, capped at 100000 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	whole transcript, capped at 100000 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	6	3	evidence hit: the shown pictures include the human-marked answer moment	66.7	4 / 6							For 66.7% of these questions, Best framework + whole transcript, capped at 100000 words showed a picture from the seconds humans marked as the answer.
314	8. Speech text	multiple-choice accuracy	Best framework + whole transcript, capped at 1500 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	whole transcript, capped at 1500 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	40.2	33 / 82	12.2	+2.5 to +22.4	yes (gain)	0.0	-3.6 to +3.2	no	Best framework + whole transcript, capped at 1500 words answered 40.2% of 82 CG-Bench dev questions correctly (2B answer model). +12.2 points vs the starting framework (confirmed). +0.0 points vs the best framework (not confirmed).
315	8. Speech text	evidence hit	Best framework + whole transcript, capped at 1500 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	whole transcript, capped at 1500 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	50.0	41 / 82							For 50.0% of these questions, Best framework + whole transcript, capped at 1500 words showed a picture from the seconds humans marked as the answer.
316	8. Speech text	open-ended correctness	Best framework + whole transcript, capped at 1500 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	whole transcript, capped at 1500 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	78	8	open-ended: the model writes its own answer (no options); AI judge marks it correct	19.2	15 / 78	6.4	+0.0 to +13.3	no				Best framework + whole transcript, capped at 1500 words wrote a correct free answer for 19.2% of 78 CG-Bench dev questions.
317	8. Speech text	multiple-choice accuracy	Best framework + whole transcript, capped at 2500 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	whole transcript, capped at 2500 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	6	3	multiple-choice accuracy (answer letter correct)	50.0	3 / 6							Best framework + whole transcript, capped at 2500 words answered 50.0% of 6 CG-Bench dev questions correctly (2B answer model).
318	8. Speech text	evidence hit	Best framework + whole transcript, capped at 2500 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	whole transcript, capped at 2500 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	6	3	evidence hit: the shown pictures include the human-marked answer moment	66.7	4 / 6							For 66.7% of these questions, Best framework + whole transcript, capped at 2500 words showed a picture from the seconds humans marked as the answer.
319	8. Speech text	multiple-choice accuracy	Best framework + best lines by word + meaning search, 120 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	best lines by word + meaning search, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	35.4	29 / 82	7.3	-4.1 to +19.7	no	-4.9	-9.9 to +0.0	no	Best framework + best lines by word + meaning search, 120 words answered 35.4% of 82 CG-Bench dev questions correctly (2B answer model). +7.3 points vs the starting framework (not confirmed). -4.9 points vs the best framework (not confirmed).
320	8. Speech text	evidence hit	Best framework + best lines by word + meaning search, 120 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	best lines by word + meaning search, 120 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	50.0	41 / 82							For 50.0% of these questions, Best framework + best lines by word + meaning search, 120 words showed a picture from the seconds humans marked as the answer.

321	8. Speech text	multiple-choice accuracy	Best framework + best lines by word + meaning search, 300 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	best lines by word + meaning search, 300 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	37.8	31 / 82	9.8	+1.1 to +20.0	yes (gain)	-2.4	-5.9 to +0.0	no	Best framework + best lines by word + meaning search, 300 words answered 37.8% of 82 CG-Bench dev questions correctly (2B answer model). +9.8 points vs the starting framework (confirmed). -2.4 points vs the best framework (not confirmed).
322	8. Speech text	evidence hit	Best framework + best lines by word + meaning search, 300 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	best lines by word + meaning search, 300 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	50.0	41 / 82							For 50.0% of these questions, Best framework + best lines by word + meaning search, 300 words showed a picture from the seconds humans marked as the answer.
323	8. Speech text	open-ended correctness	Best framework + best lines by word + meaning search, 300 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	best lines by word + meaning search, 300 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	open-ended: the model writes its own answer (no options); AI judge marks it correct	20.7	17 / 82	7.3	+0.0 to +14.1	no				Best framework + best lines by word + meaning search, 300 words wrote a correct free answer for 20.7% of 82 CG-Bench dev questions.
324	8. Speech text	multiple-choice accuracy	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	49.1	27 / 55	14.5	-1.7 to +29.8	no				Best framework + speech text answered 49.1% of 55 CG-Bench test questions correctly (2B answer model). +14.5 points vs the starting framework (not confirmed).
325	8. Speech text	evidence hit	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	evidence hit: the shown pictures include the human-marked answer moment	47.3	26 / 55							For 47.3% of these questions, Best framework + speech text showed a picture from the seconds humans marked as the answer.
326	8. Speech text	multiple-choice accuracy	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 300 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	63.3	62 / 98	7.1	+0.0 to +14.3	no	0.0	-5.1 to +6.1	no	Best framework + speech text answered 63.3% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +7.1 points vs the starting framework (not confirmed). +0.0 points vs the best framework (not confirmed).
327	8. Speech text	multiple-choice accuracy	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	60.2	59 / 98	4.1	-2.0 to +10.2	no				Best framework + speech text answered 60.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +4.1 points vs the starting framework (not confirmed).
328	8. Speech text	multiple-choice accuracy	Best framework + whole transcript, capped at 100000 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	whole transcript, capped at 100000 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	73	73	multiple-choice accuracy (answer letter correct)	58.9	43 / 73	4.1	-4.1 to +12.3	no	-2.7	-8.2 to +2.7	no	Best framework + whole transcript, capped at 100000 words answered 58.9% of 73 Video-MMMU Perception dev questions correctly (2B answer model). +4.1 points vs the starting framework (not confirmed). -2.7 points vs the best framework (not confirmed).
329	8. Speech text	multiple-choice accuracy	Best framework + whole transcript, capped at 1500 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	whole transcript, capped at 1500 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	60.2	59 / 98	4.1	-3.1 to +11.2	no	-3.1	-8.2 to +2.0	no	Best framework + whole transcript, capped at 1500 words answered 60.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +4.1 points vs the starting framework (not confirmed). -3.1 points vs the best framework (not confirmed).
330	8. Speech text	multiple-choice accuracy	Best framework + whole transcript, capped at 2500 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	whole transcript, capped at 2500 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	61.2	60 / 98	5.1	-2.0 to +12.2	no	-2.0	-8.2 to +3.1	no	Best framework + whole transcript, capped at 2500 words answered 61.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +5.1 points vs the starting framework (not confirmed). -2.0 points vs the best framework (not confirmed).

331	8. Speech text	multiple-choice accuracy	Best framework + best lines by word + meaning search, 120 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	best lines by word + meaning search, 120 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	58.2	57 / 98	2.0	-6.1 to +10.2	no	-5.1	-10.2 to +0.0	no	Best framework + best lines by word + meaning search, 120 words answered 58.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +2.0 points vs the starting framework (not confirmed). -5.1 points vs the best framework (not confirmed).
332	8. Speech text	multiple-choice accuracy	Best framework + best lines by word + meaning search, 300 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	best lines by word + meaning search, 300 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	62.2	61 / 98	6.1	-1.0 to +14.3	no	-1.0	-5.1 to +3.1	no	Best framework + best lines by word + meaning search, 300 words answered 62.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +6.1 points vs the starting framework (not confirmed). -1.0 points vs the best framework (not confirmed).
333	8. Speech text	multiple-choice accuracy	Best framework + BM25 excerpt, 120 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	33.7	33 / 98	1.0	-3.1 to +5.1	no	2.0	-3.1 to +7.1	no	Best framework + BM25 excerpt, 120 words answered 33.7% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). +1.0 points vs the starting framework (not confirmed). +2.0 points vs the best framework (not confirmed).
334	8. Speech text	multiple-choice accuracy	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 300 words	Qwen3-VL-2B	none	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	32.7	32 / 98	0.0	-4.1 to +4.1	no	1.0	-5.1 to +7.1	no	Best framework + speech text answered 32.7% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). +0.0 points vs the starting framework (not confirmed). +1.0 points vs the best framework (not confirmed).
335	8. Speech text	multiple-choice accuracy	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	none	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	34.7	34 / 98	2.0	-3.1 to +7.1	no				Best framework + speech text answered 34.7% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). +2.0 points vs the starting framework (not confirmed).
336	8. Speech text	multiple-choice accuracy	Best framework + whole transcript, capped at 100000 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	whole transcript, capped at 100000 words	Qwen3-VL-2B	none	Video-MMMU Comprehension	lectures, understanding questions	dev	15	15	multiple-choice accuracy (answer letter correct)	20.0	3 / 15	0.0	+0.0 to +0.0	no	0.0	-20.0 to +20.0	no	Best framework + whole transcript, capped at 100000 words answered 20.0% of 15 Video-MMMU Comprehension dev questions correctly (2B answer model). +0.0 points vs the starting framework (not confirmed). +0.0 points vs the best framework (not confirmed).
337	8. Speech text	multiple-choice accuracy	Best framework + whole transcript, capped at 1500 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	whole transcript, capped at 1500 words	Qwen3-VL-2B	none	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	33.7	33 / 98	1.0	-3.1 to +6.1	no	2.0	-3.1 to +8.2	no	Best framework + whole transcript, capped at 1500 words answered 33.7% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). +1.0 points vs the starting framework (not confirmed). +2.0 points vs the best framework (not confirmed).
338	8. Speech text	multiple-choice accuracy	Best framework + whole transcript, capped at 2500 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	whole transcript, capped at 2500 words	Qwen3-VL-2B	none	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	32.7	32 / 98	0.0	-4.1 to +4.1	no	1.0	-5.1 to +7.1	no	Best framework + whole transcript, capped at 2500 words answered 32.7% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). +0.0 points vs the starting framework (not confirmed). +1.0 points vs the best framework (not confirmed).
339	8. Speech text	multiple-choice accuracy	Best framework + best lines by word + meaning search, 120 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	best lines by word + meaning search, 120 words	Qwen3-VL-2B	none	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	32.7	32 / 98	0.0	-4.1 to +4.1	no	1.0	-4.1 to +6.1	no	Best framework + best lines by word + meaning search, 120 words answered 32.7% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). +0.0 points vs the starting framework (not confirmed). +1.0 points vs the best framework (not confirmed).
340	8. Speech text	multiple-choice accuracy	Best framework + best lines by word + meaning search, 300 words	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	best lines by word + meaning search, 300 words	Qwen3-VL-2B	none	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	33.7	33 / 98	1.0	-2.0 to +5.1	no	2.0	-3.1 to +8.2	no	Best framework + best lines by word + meaning search, 300 words answered 33.7% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). +1.0 points vs the starting framework (not confirmed). +2.0 points vs the best framework (not confirmed).

341	8. Speech text	multiple-choice accuracy	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	none	Video-MMMU Comprehension	lectures, understanding questions	test	202	202	multiple-choice accuracy (answer letter correct)	31.7	64 / 202	1.5	-4.0 to +6.9	no				Best framework + speech text answered 31.7% of 202 Video-MMMU Comprehension test questions correctly (2B answer model). +1.5 points vs the starting framework (not confirmed).
342	8. Speech text	multiple-choice accuracy	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	none	Video-MMMU Perception	lectures, screen-reading questions	test	202	202	multiple-choice accuracy (answer letter correct)	49.0	99 / 202	2.0	-4.0 to +7.4	no				Best framework + speech text answered 49.0% of 202 Video-MMMU Perception test questions correctly (2B answer model). +2.0 points vs the starting framework (not confirmed).
343	9. Larger answer model (4B)	multiple-choice accuracy	Best framework	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-4B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	36.6	30 / 82	0.0	-11.5 to +11.1	no				Best framework answered 36.6% of 82 CG-Bench dev questions correctly (4B answer model). +0.0 points vs the starting framework (not confirmed).
344	9. Larger answer model (4B)	evidence hit	Best framework	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-4B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	50.0	41 / 82							For 50.0% of these questions, Best framework showed a picture from the seconds humans marked as the answer.
345	9. Larger answer model (4B)	multiple-choice accuracy	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	40.2	33 / 82	3.7	-5.7 to +12.6	no				Best framework + speech text answered 40.2% of 82 CG-Bench dev questions correctly (4B answer model). +3.7 points vs the starting framework (not confirmed).
346	9. Larger answer model (4B)	evidence hit	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	50.0	41 / 82							For 50.0% of these questions, Best framework + speech text showed a picture from the seconds humans marked as the answer.
347	9. Larger answer model (4B)	multiple-choice accuracy	Starting framework (v2 design)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-4B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	36.6	30 / 82				0.0	-11.1 to +11.5	no	Starting framework (v2 design) answered 36.6% of 82 CG-Bench dev questions correctly (4B answer model). +0.0 points vs the best framework (not confirmed).
348	9. Larger answer model (4B)	evidence hit	Starting framework (v2 design)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-4B	none	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	24.4	20 / 82							For 24.4% of these questions, Starting framework (v2 design) showed a picture from the seconds humans marked as the answer.
349	9. Larger answer model (4B)	multiple-choice accuracy	Standard rule: dense MobileCLIP top-4	none	dense whole-video search, top 4	top pictures by match to the question	4	none	Qwen3-VL-4B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	40.0	22 / 55	1.8	-12.1 to +14.5	no	-1.8	-16.7 to +13.5	no	Standard rule: dense MobileCLIP top-4 answered 40.0% of 55 CG-Bench test questions correctly (4B answer model). +1.8 points vs the starting framework (not confirmed). -1.8 points vs the best framework (not confirmed).
350	9. Larger answer model (4B)	evidence hit	Standard rule: dense MobileCLIP top-4	none	dense whole-video search, top 4	top pictures by match to the question	4	none	Qwen3-VL-4B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	evidence hit: the shown pictures include the human-marked answer moment	58.2	32 / 55							For 58.2% of these questions, Standard rule: dense MobileCLIP top-4 showed a picture from the seconds humans marked as the answer.
351	9. Larger answer model (4B)	multiple-choice accuracy	Best framework	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-4B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	41.8	23 / 55	3.6	-6.9 to +16.7	no				Best framework answered 41.8% of 55 CG-Bench test questions correctly (4B answer model). +3.6 points vs the starting framework (not confirmed).
352	9. Larger answer model (4B)	evidence hit	Best framework	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-4B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	evidence hit: the shown pictures include the human-marked answer moment	47.3	26 / 55							For 47.3% of these questions, Best framework showed a picture from the seconds humans marked as the answer.

353	9. Larger answer model (4B)	multiple-choice accuracy	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	49.1	27 / 55	10.9	-1.9 to +24.1	no				Best framework + speech text answered 49.1% of 55 CG-Bench test questions correctly (4B answer model). +10.9 points vs the starting framework (not confirmed).
354	9. Larger answer model (4B)	evidence hit	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	evidence hit: the shown pictures include the human-marked answer moment	47.3	26 / 55							For 47.3% of these questions, Best framework + speech text showed a picture from the seconds humans marked as the answer.
355	9. Larger answer model (4B)	multiple-choice accuracy	Standard rule: 4 evenly spaced pictures	none	4 evenly spaced pictures	top pictures by match to the question	4	none	Qwen3-VL-4B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	27.3	15 / 55	-10.9	-21.8 to +0.0	no	-14.5	-23.6 to -5.1	yes (loss)	Standard rule: 4 evenly spaced pictures answered 27.3% of 55 CG-Bench test questions correctly (4B answer model). -10.9 points vs the starting framework (not confirmed). -14.5 points vs the best framework (confirmed).
356	9. Larger answer model (4B)	evidence hit	Standard rule: 4 evenly spaced pictures	none	4 evenly spaced pictures	top pictures by match to the question	4	none	Qwen3-VL-4B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	evidence hit: the shown pictures include the human-marked answer moment	5.5	3 / 55							For 5.5% of these questions, Standard rule: 4 evenly spaced pictures showed a picture from the seconds humans marked as the answer.
357	9. Larger answer model (4B)	multiple-choice accuracy	Starting framework (v2 design)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-4B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	38.2	21 / 55				-3.6	-16.7 to +6.9	no	Starting framework (v2 design) answered 38.2% of 55 CG-Bench test questions correctly (4B answer model). -3.6 points vs the best framework (not confirmed).
358	9. Larger answer model (4B)	evidence hit	Starting framework (v2 design)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-4B	none	CG-Bench	long everyday videos (10-80 min)	test	55	6	evidence hit: the shown pictures include the human-marked answer moment	34.5	19 / 55							For 34.5% of these questions, Starting framework (v2 design) showed a picture from the seconds humans marked as the answer.
359	9. Larger answer model (4B)	multiple-choice accuracy	Best framework	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-4B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	65.3	64 / 98	3.1	-5.1 to +11.2	no				Best framework answered 65.3% of 98 Video-MMMU Perception dev questions correctly (4B answer model). +3.1 points vs the starting framework (not confirmed).
360	9. Larger answer model (4B)	multiple-choice accuracy	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	68.4	67 / 98	6.1	-2.0 to +14.3	no				Best framework + speech text answered 68.4% of 98 Video-MMMU Perception dev questions correctly (4B answer model). +6.1 points vs the starting framework (not confirmed).
361	9. Larger answer model (4B)	multiple-choice accuracy	Starting framework (v2 design)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-4B	none	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	62.2	61 / 98				-3.1	-11.2 to +5.1	no	Starting framework (v2 design) answered 62.2% of 98 Video-MMMU Perception dev questions correctly (4B answer model). -3.1 points vs the best framework (not confirmed).
362	9. Larger answer model (4B)	multiple-choice accuracy	Best framework	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-4B	none	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	39.8	39 / 98	2.0	-3.1 to +7.1	no				Best framework answered 39.8% of 98 Video-MMMU Comprehension dev questions correctly (4B answer model). +2.0 points vs the starting framework (not confirmed).
363	9. Larger answer model (4B)	multiple-choice accuracy	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	none	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	38.8	38 / 98	1.0	-5.1 to +8.2	no				Best framework + speech text answered 38.8% of 98 Video-MMMU Comprehension dev questions correctly (4B answer model). +1.0 points vs the starting framework (not confirmed).
364	9. Larger answer model (4B)	multiple-choice accuracy	Starting framework (v2 design)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-4B	none	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	37.8	37 / 98				-2.0	-7.1 to +3.1	no	Starting framework (v2 design) answered 37.8% of 98 Video-MMMU Comprehension dev questions correctly (4B answer model). -2.0 points vs the best framework (not confirmed).

365	9. Larger answer model (4B)	multiple-choice accuracy	Best framework	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-4B	none	Video-MMMU Comprehension	lectures, understanding questions	test	202	202	multiple-choice accuracy (answer letter correct)	45.5	92 / 202	2.5	-3.0 to +7.9	no				Best framework answered 45.5% of 202 Video-MMMU Comprehension test questions correctly (4B answer model). +2.5 points vs the starting framework (not confirmed).
366	9. Larger answer model (4B)	multiple-choice accuracy	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	none	Video-MMMU Comprehension	lectures, understanding questions	test	202	202	multiple-choice accuracy (answer letter correct)	45.0	91 / 202	2.0	-3.5 to +7.9	no				Best framework + speech text answered 45.0% of 202 Video-MMMU Comprehension test questions correctly (4B answer model). +2.0 points vs the starting framework (not confirmed).
367	9. Larger answer model (4B)	multiple-choice accuracy	Starting framework (v2 design)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-4B	none	Video-MMMU Comprehension	lectures, understanding questions	test	202	202	multiple-choice accuracy (answer letter correct)	43.1	87 / 202				-2.5	-7.9 to +3.0	no	Starting framework (v2 design) answered 43.1% of 202 Video-MMMU Comprehension test questions correctly (4B answer model). -2.5 points vs the best framework (not confirmed).
368	9. Larger answer model (4B)	multiple-choice accuracy	Standard rule: dense MobileCLIP top-4	none	dense whole-video search, top 4	top pictures by match to the question	4	none	Qwen3-VL-4B	none	Video-MMMU Perception	lectures, screen-reading questions	test	188	188	multiple-choice accuracy (answer letter correct)	54.8	103 / 188	-4.8	-11.7 to +2.7	no	-8.0	-13.8 to -2.1	yes (loss)	Standard rule: dense MobileCLIP top-4 answered 54.8% of 188 Video-MMMU Perception test questions correctly (4B answer model). -4.8 points vs the starting framework (not confirmed). -8.0 points vs the best framework (confirmed).
369	9. Larger answer model (4B)	multiple-choice accuracy	Best framework	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-4B	none	Video-MMMU Perception	lectures, screen-reading questions	test	202	202	multiple-choice accuracy (answer letter correct)	62.9	127 / 202	2.0	-4.5 to +8.9	no				Best framework answered 62.9% of 202 Video-MMMU Perception test questions correctly (4B answer model). +2.0 points vs the starting framework (not confirmed).
370	9. Larger answer model (4B)	multiple-choice accuracy	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	none	Video-MMMU Perception	lectures, screen-reading questions	test	202	202	multiple-choice accuracy (answer letter correct)	67.8	137 / 202	6.9	+0.5 to +13.9	yes (gain)				Best framework + speech text answered 67.8% of 202 Video-MMMU Perception test questions correctly (4B answer model). +6.9 points vs the starting framework (confirmed).
371	9. Larger answer model (4B)	multiple-choice accuracy	Starting framework (v2 design)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-4B	none	Video-MMMU Perception	lectures, screen-reading questions	test	202	202	multiple-choice accuracy (answer letter correct)	60.9	123 / 202				-2.0	-8.9 to +4.5	no	Starting framework (v2 design) answered 60.9% of 202 Video-MMMU Perception test questions correctly (4B answer model). -2.0 points vs the best framework (not confirmed).
372	10. Full combination grid	multiple-choice accuracy	Best framework + Head B	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-4B	Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	40.2	33 / 82	3.7	-4.1 to +12.2	no	3.7	-2.1 to +10.1	no	Best framework + Head B answered 40.2% of 82 CG-Bench dev questions correctly (4B answer model). +3.7 points vs the starting framework (not confirmed). +3.7 points vs the best framework (not confirmed).
373	10. Full combination grid	evidence hit	Best framework + Head B	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-4B	Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	48.8	40 / 82							For 48.8% of these questions, Best framework + Head B showed a picture from the seconds humans marked as the answer.
374	10. Full combination grid	multiple-choice accuracy	Best framework + Head B + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	37.8	31 / 82	9.8	+1.1 to +20.0	yes (gain)	0.0	-4.3 to +5.3	no	Best framework + Head B + speech text answered 37.8% of 82 CG-Bench dev questions correctly (2B answer model). +9.8 points vs the starting framework (confirmed). +0.0 points vs the best framework (not confirmed).
375	10. Full combination grid	evidence hit	Best framework + Head B + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	48.8	40 / 82							For 48.8% of these questions, Best framework + Head B + speech text showed a picture from the seconds humans marked as the answer.
376	10. Full combination grid	multiple-choice accuracy	Best framework + Head B + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	46.3	38 / 82	9.8	+0.0 to +20.2	no	6.1	-1.2 to +14.5	no	Best framework + Head B + speech text answered 46.3% of 82 CG-Bench dev questions correctly (4B answer model). +9.8 points vs the starting framework (not confirmed). +6.1 points vs the best framework (not confirmed).

377	10. Full combination grid	evidence hit	Best framework + Head B + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	48.8	40 / 82							For 48.8% of these questions, Best framework + Head B + speech text showed a picture from the seconds humans marked as the answer.
378	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + Head B	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-4B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	47.6	39 / 82	11.0	+0.0 to +21.3	no	11.0	+5.6 to +17.1	yes (gain)	Best framework + Head A + Head B answered 47.6% of 82 CG-Bench dev questions correctly (4B answer model). +11.0 points vs the starting framework (not confirmed). +11.0 points vs the best framework (confirmed).
379	10. Full combination grid	evidence hit	Best framework + Head A + Head B	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-4B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	47.6	39 / 82							For 47.6% of these questions, Best framework + Head A + Head B showed a picture from the seconds humans marked as the answer.
380	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + Head B + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	36.6	30 / 82	8.5	-1.0 to +19.7	no	-1.2	-8.0 to +8.0	no	Best framework + Head A + Head B + speech text answered 36.6% of 82 CG-Bench dev questions correctly (2B answer model). +8.5 points vs the starting framework (not confirmed). -1.2 points vs the best framework (not confirmed).
381	10. Full combination grid	evidence hit	Best framework + Head A + Head B + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	47.6	39 / 82							For 47.6% of these questions, Best framework + Head A + Head B + speech text showed a picture from the seconds humans marked as the answer.
382	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + Head B + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	50.0	41 / 82	13.4	+4.5 to +24.1	yes (gain)	9.8	+3.7 to +15.9	yes (gain)	Best framework + Head A + Head B + speech text answered 50.0% of 82 CG-Bench dev questions correctly (4B answer model). +13.4 points vs the starting framework (confirmed). +9.8 points vs the best framework (confirmed).
383	10. Full combination grid	evidence hit	Best framework + Head A + Head B + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	47.6	39 / 82							For 47.6% of these questions, Best framework + Head A + Head B + speech text showed a picture from the seconds humans marked as the answer.
384	10. Full combination grid	multiple-choice accuracy	Best framework + Head A	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-4B	Head A	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	45.1	37 / 82	8.5	-1.5 to +17.8	no	8.5	+4.6 to +12.9	yes (gain)	Best framework + Head A answered 45.1% of 82 CG-Bench dev questions correctly (4B answer model). +8.5 points vs the starting framework (not confirmed). +8.5 points vs the best framework (confirmed).
385	10. Full combination grid	evidence hit	Best framework + Head A	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-4B	Head A	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	45.1	37 / 82							For 45.1% of these questions, Best framework + Head A showed a picture from the seconds humans marked as the answer.
386	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head A	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	37.8	31 / 82	9.8	+1.1 to +20.3	yes (gain)	0.0	-3.1 to +4.1	no	Best framework + Head A + speech text answered 37.8% of 82 CG-Bench dev questions correctly (2B answer model). +9.8 points vs the starting framework (confirmed). +0.0 points vs the best framework (not confirmed).
387	10. Full combination grid	evidence hit	Best framework + Head A + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head A	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	45.1	37 / 82							For 45.1% of these questions, Best framework + Head A + speech text showed a picture from the seconds humans marked as the answer.

388	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head A	CG-Bench	long everyday videos (10-80 min)	dev	82	8	multiple-choice accuracy (answer letter correct)	51.2	42 / 82	14.6	+2.7 to +24.7	yes (gain)	11.0	+2.7 to +18.2	yes (gain)	Best framework + Head A + speech text answered 51.2% of 82 CG-Bench dev questions correctly (4B answer model). +14.6 points vs the starting framework (confirmed). +11.0 points vs the best framework (confirmed).
389	10. Full combination grid	evidence hit	Best framework + Head A + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head A	CG-Bench	long everyday videos (10-80 min)	dev	82	8	evidence hit: the shown pictures include the human-marked answer moment	45.1	37 / 82							For 45.1% of these questions, Best framework + Head A + speech text showed a picture from the seconds humans marked as the answer.
390	10. Full combination grid	multiple-choice accuracy	Best framework + Head B	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-4B	Head B	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	38.2	21 / 55	0.0	-5.1 to +5.4	no	-3.6	-20.8 to +10.5	no	Best framework + Head B answered 38.2% of 55 CG-Bench test questions correctly (4B answer model). +0.0 points vs the starting framework (not confirmed). -3.6 points vs the best framework (not confirmed).
391	10. Full combination grid	evidence hit	Best framework + Head B	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-4B	Head B	CG-Bench	long everyday videos (10-80 min)	test	55	6	evidence hit: the shown pictures include the human-marked answer moment	50.9	28 / 55							For 50.9% of these questions, Best framework + Head B showed a picture from the seconds humans marked as the answer.
392	10. Full combination grid	multiple-choice accuracy	Best framework + Head B + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head B	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	40.0	22 / 55	5.5	-7.1 to +18.2	no	-9.1	-16.0 to -3.2	yes (loss)	Best framework + Head B + speech text answered 40.0% of 55 CG-Bench test questions correctly (2B answer model). +5.5 points vs the starting framework (not confirmed). -9.1 points vs the best framework (confirmed).
393	10. Full combination grid	evidence hit	Best framework + Head B + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head B	CG-Bench	long everyday videos (10-80 min)	test	55	6	evidence hit: the shown pictures include the human-marked answer moment	50.9	28 / 55							For 50.9% of these questions, Best framework + Head B + speech text showed a picture from the seconds humans marked as the answer.
394	10. Full combination grid	multiple-choice accuracy	Best framework + Head B + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head B	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	43.6	24 / 55	5.5	-1.8 to +12.0	no	-5.5	-22.6 to +9.6	no	Best framework + Head B + speech text answered 43.6% of 55 CG-Bench test questions correctly (4B answer model). +5.5 points vs the starting framework (not confirmed). -5.5 points vs the best framework (not confirmed).
395	10. Full combination grid	evidence hit	Best framework + Head B + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head B	CG-Bench	long everyday videos (10-80 min)	test	55	6	evidence hit: the shown pictures include the human-marked answer moment	50.9	28 / 55							For 50.9% of these questions, Best framework + Head B + speech text showed a picture from the seconds humans marked as the answer.
396	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + Head B	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-4B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	34.5	19 / 55	-3.6	-16.4 to +7.4	no	-7.3	-16.4 to +2.0	no	Best framework + Head A + Head B answered 34.5% of 55 CG-Bench test questions correctly (4B answer model). -3.6 points vs the starting framework (not confirmed). -7.3 points vs the best framework (not confirmed).
397	10. Full combination grid	evidence hit	Best framework + Head A + Head B	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-4B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	test	55	6	evidence hit: the shown pictures include the human-marked answer moment	56.4	31 / 55							For 56.4% of these questions, Best framework + Head A + Head B showed a picture from the seconds humans marked as the answer.
398	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + Head B + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	34.5	19 / 55	0.0	-10.5 to +10.0	no	-14.5	-22.9 to -6.9	yes (loss)	Best framework + Head A + Head B + speech text answered 34.5% of 55 CG-Bench test questions correctly (2B answer model). +0.0 points vs the starting framework (not confirmed). -14.5 points vs the best framework (confirmed).

399	10. Full combination grid	evidence hit	Best framework + Head A + Head B + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	test	55	6	evidence hit: the shown pictures include the human-marked answer moment	56.4	31 / 55							For 56.4% of these questions, Best framework + Head A + Head B + speech text showed a picture from the seconds humans marked as the answer.
400	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + Head B + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	45.5	25 / 55	7.3	+0.0 to +12.5	no	-3.6	-14.3 to +7.0	no	Best framework + Head A + Head B + speech text answered 45.5% of 55 CG-Bench test questions correctly (4B answer model). +7.3 points vs the starting framework (not confirmed). -3.6 points vs the best framework (not confirmed).
401	10. Full combination grid	evidence hit	Best framework + Head A + Head B + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head A, Head B	CG-Bench	long everyday videos (10-80 min)	test	55	6	evidence hit: the shown pictures include the human-marked answer moment	56.4	31 / 55							For 56.4% of these questions, Best framework + Head A + Head B + speech text showed a picture from the seconds humans marked as the answer.
402	10. Full combination grid	multiple-choice accuracy	Best framework + Head A	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-4B	Head A	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	41.8	23 / 55	3.6	-11.5 to +14.8	no	0.0	-13.3 to +15.1	no	Best framework + Head A answered 41.8% of 55 CG-Bench test questions correctly (4B answer model). +3.6 points vs the starting framework (not confirmed). +0.0 points vs the best framework (not confirmed).
403	10. Full combination grid	evidence hit	Best framework + Head A	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-4B	Head A	CG-Bench	long everyday videos (10-80 min)	test	55	6	evidence hit: the shown pictures include the human-marked answer moment	49.1	27 / 55							For 49.1% of these questions, Best framework + Head A showed a picture from the seconds humans marked as the answer.
404	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head A	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	40.0	22 / 55	5.5	-5.1 to +18.8	no	-9.1	-17.3 to +0.0	no	Best framework + Head A + speech text answered 40.0% of 55 CG-Bench test questions correctly (2B answer model). +5.5 points vs the starting framework (not confirmed). -9.1 points vs the best framework (not confirmed).
405	10. Full combination grid	evidence hit	Best framework + Head A + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head A	CG-Bench	long everyday videos (10-80 min)	test	55	6	evidence hit: the shown pictures include the human-marked answer moment	49.1	27 / 55							For 49.1% of these questions, Best framework + Head A + speech text showed a picture from the seconds humans marked as the answer.
406	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head A	CG-Bench	long everyday videos (10-80 min)	test	55	6	multiple-choice accuracy (answer letter correct)	47.3	26 / 55	9.1	+1.8 to +16.7	yes (gain)	-1.8	-13.0 to +10.5	no	Best framework + Head A + speech text answered 47.3% of 55 CG-Bench test questions correctly (4B answer model). +9.1 points vs the starting framework (confirmed). -1.8 points vs the best framework (not confirmed).
407	10. Full combination grid	evidence hit	Best framework + Head A + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head A	CG-Bench	long everyday videos (10-80 min)	test	55	6	evidence hit: the shown pictures include the human-marked answer moment	49.1	27 / 55							For 49.1% of these questions, Best framework + Head A + speech text showed a picture from the seconds humans marked as the answer.
408	10. Full combination grid	multiple-choice accuracy	Best framework + Head B	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-4B	Head B	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	64.3	63 / 98	2.0	-5.1 to +9.2	no	-1.0	-9.2 to +7.1	no	Best framework + Head B answered 64.3% of 98 Video-MMMU Perception dev questions correctly (4B answer model). +2.0 points vs the starting framework (not confirmed). -1.0 points vs the best framework (not confirmed).

409	10. Full combination grid	multiple-choice accuracy	Best framework + Head B + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head B	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	62.2	61 / 98	6.1	-2.0 to +14.3	no	2.0	-6.1 to +9.2	no	Best framework + Head B + speech text answered 62.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +6.1 points vs the starting framework (not confirmed). +2.0 points vs the best framework (not confirmed).
410	10. Full combination grid	multiple-choice accuracy	Best framework + Head B + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head B	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	67.3	66 / 98	5.1	-2.0 to +12.2	no	-1.0	-10.2 to +8.2	no	Best framework + Head B + speech text answered 67.3% of 98 Video-MMMU Perception dev questions correctly (4B answer model). +5.1 points vs the starting framework (not confirmed). -1.0 points vs the best framework (not confirmed).
411	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + Head B	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-4B	Head A, Head B	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	66.3	65 / 98	4.1	-3.1 to +12.2	no	1.0	-7.1 to +9.2	no	Best framework + Head A + Head B answered 66.3% of 98 Video-MMMU Perception dev questions correctly (4B answer model). +4.1 points vs the starting framework (not confirmed). +1.0 points vs the best framework (not confirmed).
412	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + Head B + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head A, Head B	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	60.2	59 / 98	4.1	-4.1 to +12.2	no	0.0	-7.1 to +7.1	no	Best framework + Head A + Head B + speech text answered 60.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +4.1 points vs the starting framework (not confirmed). +0.0 points vs the best framework (not confirmed).
413	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + Head B + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head A, Head B	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	70.4	69 / 98	8.2	+0.0 to +15.3	no	2.0	-6.1 to +10.2	no	Best framework + Head A + Head B + speech text answered 70.4% of 98 Video-MMMU Perception dev questions correctly (4B answer model). +8.2 points vs the starting framework (not confirmed). +2.0 points vs the best framework (not confirmed).
414	10. Full combination grid	multiple-choice accuracy	Best framework + Head A	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-4B	Head A	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	65.3	64 / 98	3.1	-5.1 to +11.2	no	0.0	-5.1 to +5.1	no	Best framework + Head A answered 65.3% of 98 Video-MMMU Perception dev questions correctly (4B answer model). +3.1 points vs the starting framework (not confirmed). +0.0 points vs the best framework (not confirmed).
415	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head A	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	58.2	57 / 98	2.0	-5.1 to +10.2	no	-2.0	-7.1 to +3.1	no	Best framework + Head A + speech text answered 58.2% of 98 Video-MMMU Perception dev questions correctly (2B answer model). +2.0 points vs the starting framework (not confirmed). -2.0 points vs the best framework (not confirmed).
416	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head A	Video-MMMU Perception	lectures, screen-reading questions	dev	98	98	multiple-choice accuracy (answer letter correct)	65.3	64 / 98	3.1	-5.1 to +11.2	no	-3.1	-8.2 to +2.0	no	Best framework + Head A + speech text answered 65.3% of 98 Video-MMMU Perception dev questions correctly (4B answer model). +3.1 points vs the starting framework (not confirmed). -3.1 points vs the best framework (not confirmed).
417	10. Full combination grid	multiple-choice accuracy	Best framework + Head B	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-4B	Head B	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	43.9	43 / 98	6.1	+0.0 to +12.2	no	4.1	-3.1 to +11.2	no	Best framework + Head B answered 43.9% of 98 Video-MMMU Comprehension dev questions correctly (4B answer model). +6.1 points vs the starting framework (not confirmed). +4.1 points vs the best framework (not confirmed).
418	10. Full combination grid	multiple-choice accuracy	Best framework + Head B + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head B	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	29.6	29 / 98	-3.1	-9.2 to +3.1	no	-5.1	-11.2 to +1.0	no	Best framework + Head B + speech text answered 29.6% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). -3.1 points vs the starting framework (not confirmed). -5.1 points vs the best framework (not confirmed).

419	10. Full combination grid	multiple-choice accuracy	Best framework + Head B + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head B	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	38.8	38 / 98	1.0	-6.1 to +9.2	no	0.0	-6.1 to +6.1	no	Best framework + Head B + speech text answered 38.8% of 98 Video-MMMU Comprehension dev questions correctly (4B answer model). +1.0 points vs the starting framework (not confirmed). +0.0 points vs the best framework (not confirmed).
420	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + Head B	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-4B	Head A, Head B	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	41.8	41 / 98	4.1	-1.0 to +10.2	no	2.0	-4.1 to +8.2	no	Best framework + Head A + Head B answered 41.8% of 98 Video-MMMU Comprehension dev questions correctly (4B answer model). +4.1 points vs the starting framework (not confirmed). +2.0 points vs the best framework (not confirmed).
421	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + Head B + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head A, Head B	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	30.6	30 / 98	-2.0	-9.2 to +5.1	no	-4.1	-10.2 to +2.0	no	Best framework + Head A + Head B + speech text answered 30.6% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). -2.0 points vs the starting framework (not confirmed). -4.1 points vs the best framework (not confirmed).
422	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + Head B + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head A, Head B	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	40.8	40 / 98	3.1	-3.1 to +10.2	no	2.0	-3.1 to +7.1	no	Best framework + Head A + Head B + speech text answered 40.8% of 98 Video-MMMU Comprehension dev questions correctly (4B answer model). +3.1 points vs the starting framework (not confirmed). +2.0 points vs the best framework (not confirmed).
423	10. Full combination grid	multiple-choice accuracy	Best framework + Head A	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-4B	Head A	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	36.7	36 / 98	-1.0	-7.1 to +5.1	no	-3.1	-7.1 to +1.0	no	Best framework + Head A answered 36.7% of 98 Video-MMMU Comprehension dev questions correctly (4B answer model). -1.0 points vs the starting framework (not confirmed). -3.1 points vs the best framework (not confirmed).
424	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head A	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	29.6	29 / 98	-3.1	-8.2 to +2.0	no	-5.1	-10.2 to +0.0	no	Best framework + Head A + speech text answered 29.6% of 98 Video-MMMU Comprehension dev questions correctly (2B answer model). -3.1 points vs the starting framework (not confirmed). -5.1 points vs the best framework (not confirmed).
425	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head A	Video-MMMU Comprehension	lectures, understanding questions	dev	98	98	multiple-choice accuracy (answer letter correct)	37.8	37 / 98	0.0	-8.2 to +8.2	no	-1.0	-7.1 to +5.1	no	Best framework + Head A + speech text answered 37.8% of 98 Video-MMMU Comprehension dev questions correctly (4B answer model). +0.0 points vs the starting framework (not confirmed). -1.0 points vs the best framework (not confirmed).
426	10. Full combination grid	multiple-choice accuracy	Best framework + Head B	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-4B	Head B	Video-MMMU Comprehension	lectures, understanding questions	test	202	202	multiple-choice accuracy (answer letter correct)	48.5	98 / 202	5.4	-0.5 to +10.9	no	3.0	-1.5 to +7.9	no	Best framework + Head B answered 48.5% of 202 Video-MMMU Comprehension test questions correctly (4B answer model). +5.4 points vs the starting framework (not confirmed). +3.0 points vs the best framework (not confirmed).
427	10. Full combination grid	multiple-choice accuracy	Best framework + Head B + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head B	Video-MMMU Comprehension	lectures, understanding questions	test	202	202	multiple-choice accuracy (answer letter correct)	34.7	70 / 202	4.5	-1.0 to +10.4	no	3.0	-1.5 to +7.4	no	Best framework + Head B + speech text answered 34.7% of 202 Video-MMMU Comprehension test questions correctly (2B answer model). +4.5 points vs the starting framework (not confirmed). +3.0 points vs the best framework (not confirmed).
428	10. Full combination grid	multiple-choice accuracy	Best framework + Head B + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head B	Video-MMMU Comprehension	lectures, understanding questions	test	202	202	multiple-choice accuracy (answer letter correct)	45.5	92 / 202	2.5	-3.5 to +8.4	no	0.5	-4.0 to +5.0	no	Best framework + Head B + speech text answered 45.5% of 202 Video-MMMU Comprehension test questions correctly (4B answer model). +2.5 points vs the starting framework (not confirmed). +0.5 points vs the best framework (not confirmed).

429	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + Head B	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-4B	Head A, Head B	Video-MMMU Comprehension	lectures, understanding questions	test	202	202	multiple-choice accuracy (answer letter correct)	49.0	99 / 202	5.9	+1.5 to +10.9	yes (gain)	3.5	-1.5 to +8.4	no	Best framework + Head A + Head B answered 49.0% of 202 Video-MMMU Comprehension test questions correctly (4B answer model). +5.9 points vs the starting framework (confirmed). +3.5 points vs the best framework (not confirmed).
430	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + Head B + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head A, Head B	Video-MMMU Comprehension	lectures, understanding questions	test	202	202	multiple-choice accuracy (answer letter correct)	32.7	66 / 202	2.5	-3.0 to +8.4	no	1.0	-4.0 to +5.9	no	Best framework + Head A + Head B + speech text answered 32.7% of 202 Video-MMMU Comprehension test questions correctly (2B answer model). +2.5 points vs the starting framework (not confirmed). +1.0 points vs the best framework (not confirmed).
431	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + Head B + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head A, Head B	Video-MMMU Comprehension	lectures, understanding questions	test	202	202	multiple-choice accuracy (answer letter correct)	45.5	92 / 202	2.5	-3.0 to +7.9	no	0.5	-5.0 to +5.4	no	Best framework + Head A + Head B + speech text answered 45.5% of 202 Video-MMMU Comprehension test questions correctly (4B answer model). +2.5 points vs the starting framework (not confirmed). +0.5 points vs the best framework (not confirmed).
432	10. Full combination grid	multiple-choice accuracy	Best framework + Head A	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-4B	Head A	Video-MMMU Comprehension	lectures, understanding questions	test	202	202	multiple-choice accuracy (answer letter correct)	45.5	92 / 202	2.5	-2.5 to +7.4	no	0.0	-3.5 to +3.5	no	Best framework + Head A answered 45.5% of 202 Video-MMMU Comprehension test questions correctly (4B answer model). +2.5 points vs the starting framework (not confirmed). +0.0 points vs the best framework (not confirmed).
433	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head A	Video-MMMU Comprehension	lectures, understanding questions	test	202	202	multiple-choice accuracy (answer letter correct)	33.2	67 / 202	3.0	-2.5 to +8.9	no	1.5	-2.5 to +5.4	no	Best framework + Head A + speech text answered 33.2% of 202 Video-MMMU Comprehension test questions correctly (2B answer model). +3.0 points vs the starting framework (not confirmed). +1.5 points vs the best framework (not confirmed).
434	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head A	Video-MMMU Comprehension	lectures, understanding questions	test	202	202	multiple-choice accuracy (answer letter correct)	47.0	95 / 202	4.0	-1.5 to +9.9	no	2.0	-2.0 to +5.9	no	Best framework + Head A + speech text answered 47.0% of 202 Video-MMMU Comprehension test questions correctly (4B answer model). +4.0 points vs the starting framework (not confirmed). +2.0 points vs the best framework (not confirmed).
435	10. Full combination grid	multiple-choice accuracy	Best framework + Head B	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-4B	Head B	Video-MMMU Perception	lectures, screen-reading questions	test	202	202	multiple-choice accuracy (answer letter correct)	64.4	130 / 202	3.5	-2.0 to +8.9	no	1.5	-4.5 to +7.4	no	Best framework + Head B answered 64.4% of 202 Video-MMMU Perception test questions correctly (4B answer model). +3.5 points vs the starting framework (not confirmed). +1.5 points vs the best framework (not confirmed).
436	10. Full combination grid	multiple-choice accuracy	Best framework + Head B + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head B	Video-MMMU Perception	lectures, screen-reading questions	test	202	202	multiple-choice accuracy (answer letter correct)	47.5	96 / 202	0.5	-4.5 to +5.4	no	-1.5	-6.9 to +3.5	no	Best framework + Head B + speech text answered 47.5% of 202 Video-MMMU Perception test questions correctly (2B answer model). +0.5 points vs the starting framework (not confirmed). -1.5 points vs the best framework (not confirmed).
437	10. Full combination grid	multiple-choice accuracy	Best framework + Head B + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head B	Video-MMMU Perception	lectures, screen-reading questions	test	202	202	multiple-choice accuracy (answer letter correct)	64.9	131 / 202	4.0	-2.0 to +10.4	no	-3.0	-8.4 to +2.5	no	Best framework + Head B + speech text answered 64.9% of 202 Video-MMMU Perception test questions correctly (4B answer model). +4.0 points vs the starting framework (not confirmed). -3.0 points vs the best framework (not confirmed).
438	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + Head B	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-4B	Head A, Head B	Video-MMMU Perception	lectures, screen-reading questions	test	202	202	multiple-choice accuracy (answer letter correct)	66.3	134 / 202	5.4	-0.5 to +11.9	no	3.5	-2.5 to +8.9	no	Best framework + Head A + Head B answered 66.3% of 202 Video-MMMU Perception test questions correctly (4B answer model). +5.4 points vs the starting framework (not confirmed). +3.5 points vs the best framework (not confirmed).

439	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + Head B + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head A, Head B	Video-MMMU Perception	lectures, screen-reading questions	test	202	202	multiple-choice accuracy (answer letter correct)	51.0	103 / 202	4.0	-1.5 to +9.4	no	2.0	-3.0 to +6.9	no	Best framework + Head A + Head B + speech text answered 51.0% of 202 Video-MMMU Perception test questions correctly (2B answer model). +4.0 points vs the starting framework (not confirmed). +2.0 points vs the best framework (not confirmed).
440	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + Head B + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head A, Head B	Video-MMMU Perception	lectures, screen-reading questions	test	202	202	multiple-choice accuracy (answer letter correct)	66.8	135 / 202	5.9	-0.5 to +12.4	no	-1.0	-6.4 to +4.5	no	Best framework + Head A + Head B + speech text answered 66.8% of 202 Video-MMMU Perception test questions correctly (4B answer model). +5.9 points vs the starting framework (not confirmed). -1.0 points vs the best framework (not confirmed).
441	10. Full combination grid	multiple-choice accuracy	Best framework + Head A	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-4B	Head A	Video-MMMU Perception	lectures, screen-reading questions	test	202	202	multiple-choice accuracy (answer letter correct)	65.8	133 / 202	5.0	-1.0 to +11.4	no	3.0	-2.0 to +7.9	no	Best framework + Head A answered 65.8% of 202 Video-MMMU Perception test questions correctly (4B answer model). +5.0 points vs the starting framework (not confirmed). +3.0 points vs the best framework (not confirmed).
442	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head A	Video-MMMU Perception	lectures, screen-reading questions	test	202	202	multiple-choice accuracy (answer letter correct)	48.0	97 / 202	1.0	-5.0 to +6.9	no	-1.0	-5.4 to +3.5	no	Best framework + Head A + speech text answered 48.0% of 202 Video-MMMU Perception test questions correctly (2B answer model). +1.0 points vs the starting framework (not confirmed). -1.0 points vs the best framework (not confirmed).
443	10. Full combination grid	multiple-choice accuracy	Best framework + Head A + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head A	Video-MMMU Perception	lectures, screen-reading questions	test	202	202	multiple-choice accuracy (answer letter correct)	66.8	135 / 202	5.9	+0.0 to +12.4	no	-1.0	-5.0 to +3.0	no	Best framework + Head A + speech text answered 66.8% of 202 Video-MMMU Perception test questions correctly (4B answer model). +5.9 points vs the starting framework (not confirmed). -1.0 points vs the best framework (not confirmed).
444	11. Pooled summaries	multiple-choice accuracy	Standard rule: dense MobileCLIP top-4	none	dense whole-video search, top 4	top pictures by match to the question	4	none	Qwen3-VL-2B	none	Video-MMMU (both tracks)	all lecture test questions	test	404	404	multiple-choice accuracy (answer letter correct)	36.1	146 / 404	-2.5	-6.4 to +1.7	no	-4.0	-7.4 to -0.5	yes (loss)	Standard rule: dense MobileCLIP top-4 answered 36.1% of 404 Video-MMMU (both tracks) test questions correctly (2B answer model). -2.5 points vs the starting framework (not confirmed). -4.0 points vs the best framework (confirmed).
445	11. Pooled summaries	multiple-choice accuracy	Best framework + Head B	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-2B	Head B	Video-MMMU (both tracks)	all lecture test questions	test	404	404	multiple-choice accuracy (answer letter correct)	40.6	164 / 404	2.0	-1.5 to +5.7	no	0.5	-2.7 to +3.7	no	Best framework + Head B answered 40.6% of 404 Video-MMMU (both tracks) test questions correctly (2B answer model). +2.0 points vs the starting framework (not confirmed). +0.5 points vs the best framework (not confirmed).
446	11. Pooled summaries	multiple-choice accuracy	Best framework + Head B	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-4B	Head B	Video-MMMU (both tracks)	all lecture test questions	test	404	404	multiple-choice accuracy (answer letter correct)	56.4	228 / 404	4.5	+0.5 to +8.4	yes (gain)	2.2	-1.5 to +5.9	no	Best framework + Head B answered 56.4% of 404 Video-MMMU (both tracks) test questions correctly (4B answer model). +4.5 points vs the starting framework (confirmed). +2.2 points vs the best framework (not confirmed).
447	11. Pooled summaries	multiple-choice accuracy	Best framework + Head B + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head B	Video-MMMU (both tracks)	all lecture test questions	test	404	404	multiple-choice accuracy (answer letter correct)	41.1	166 / 404	2.5	-1.2 to +6.2	no	0.7	-2.7 to +4.2	no	Best framework + Head B + speech text answered 41.1% of 404 Video-MMMU (both tracks) test questions correctly (2B answer model). +2.5 points vs the starting framework (not confirmed). +0.7 points vs the best framework (not confirmed).
448	11. Pooled summaries	multiple-choice accuracy	Best framework + Head B + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head B	Video-MMMU (both tracks)	all lecture test questions	test	404	404	multiple-choice accuracy (answer letter correct)	55.2	223 / 404	3.2	-1.2 to +7.7	no	-1.2	-4.7 to +2.5	no	Best framework + Head B + speech text answered 55.2% of 404 Video-MMMU (both tracks) test questions correctly (4B answer model). +3.2 points vs the starting framework (not confirmed). -1.2 points vs the best framework (not confirmed).
449	11. Pooled summaries	multiple-choice accuracy	Best framework	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	none	Video-MMMU (both tracks)	all lecture test questions	test	404	404	multiple-choice accuracy (answer letter correct)	40.1	162 / 404	1.5	-2.5 to +5.4	no				Best framework answered 40.1% of 404 Video-MMMU (both tracks) test questions correctly (2B answer model). +1.5 points vs the starting framework (not confirmed).

450	11. Pooled summaries	multiple-choice accuracy	Best framework	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-4B	none	Video-MMMU (both tracks)	all lecture test questions	test	404	404	multiple-choice accuracy (answer letter correct)	54.2	219 / 404	2.2	-2.0 to +6.4	no				Best framework answered 54.2% of 404 Video-MMMU (both tracks) test questions correctly (4B answer model). +2.2 points vs the starting framework (not confirmed).
451	11. Pooled summaries	multiple-choice accuracy	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	none	Video-MMMU (both tracks)	all lecture test questions	test	404	404	multiple-choice accuracy (answer letter correct)	40.3	163 / 404	1.7	-2.2 to +5.7	no				Best framework + speech text answered 40.3% of 404 Video-MMMU (both tracks) test questions correctly (2B answer model). +1.7 points vs the starting framework (not confirmed).
452	11. Pooled summaries	multiple-choice accuracy	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	none	Video-MMMU (both tracks)	all lecture test questions	test	404	404	multiple-choice accuracy (answer letter correct)	56.4	228 / 404	4.5	+0.2 to +8.9	yes (gain)				Best framework + speech text answered 56.4% of 404 Video-MMMU (both tracks) test questions correctly (4B answer model). +4.5 points vs the starting framework (confirmed).
453	11. Pooled summaries	multiple-choice accuracy	No pictures (question and options only)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	no pictures	0	none	Qwen3-VL-2B	none	Video-MMMU (both tracks)	all lecture test questions	test	404	404	multiple-choice accuracy (answer letter correct)	22.5	91 / 404	-16.1	-21.0 to -11.4	yes (loss)	-17.6	-22.8 to -12.4	yes (loss)	No pictures (question and options only) answered 22.5% of 404 Video-MMMU (both tracks) test questions correctly (2B answer model). -16.1 points vs the starting framework (confirmed). -17.6 points vs the best framework (confirmed).
454	11. Pooled summaries	multiple-choice accuracy	Best framework + Head A + Head B	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-2B	Head A, Head B	Video-MMMU (both tracks)	all lecture test questions	test	404	404	multiple-choice accuracy (answer letter correct)	42.3	171 / 404	3.7	+0.0 to +7.4	no	2.2	-0.7 to +5.4	no	Best framework + Head A + Head B answered 42.3% of 404 Video-MMMU (both tracks) test questions correctly (2B answer model). +3.7 points vs the starting framework (not confirmed). +2.2 points vs the best framework (not confirmed).
455	11. Pooled summaries	multiple-choice accuracy	Best framework + Head A + Head B	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-4B	Head A, Head B	Video-MMMU (both tracks)	all lecture test questions	test	404	404	multiple-choice accuracy (answer letter correct)	57.7	233 / 404	5.7	+1.7 to +9.7	yes (gain)	3.5	-0.5 to +7.2	no	Best framework + Head A + Head B answered 57.7% of 404 Video-MMMU (both tracks) test questions correctly (4B answer model). +5.7 points vs the starting framework (confirmed). +3.5 points vs the best framework (not confirmed).
456	11. Pooled summaries	multiple-choice accuracy	Best framework + Head A + Head B + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head A, Head B	Video-MMMU (both tracks)	all lecture test questions	test	404	404	multiple-choice accuracy (answer letter correct)	41.8	169 / 404	3.2	-0.7 to +6.9	no	1.5	-2.0 to +5.0	no	Best framework + Head A + Head B + speech text answered 41.8% of 404 Video-MMMU (both tracks) test questions correctly (2B answer model). +3.2 points vs the starting framework (not confirmed). +1.5 points vs the best framework (not confirmed).
457	11. Pooled summaries	multiple-choice accuracy	Best framework + Head A + Head B + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head A, Head B	Video-MMMU (both tracks)	all lecture test questions	test	404	404	multiple-choice accuracy (answer letter correct)	56.2	227 / 404	4.2	+0.0 to +8.2	no	-0.2	-4.0 to +3.5	no	Best framework + Head A + Head B + speech text answered 56.2% of 404 Video-MMMU (both tracks) test questions correctly (4B answer model). +4.2 points vs the starting framework (not confirmed). -0.2 points vs the best framework (not confirmed).
458	11. Pooled summaries	multiple-choice accuracy	Best framework + Head A	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	Head A	Video-MMMU (both tracks)	all lecture test questions	test	404	404	multiple-choice accuracy (answer letter correct)	39.4	159 / 404	0.7	-3.5 to +5.0	no	-0.7	-3.5 to +2.2	no	Best framework + Head A answered 39.4% of 404 Video-MMMU (both tracks) test questions correctly (2B answer model). +0.7 points vs the starting framework (not confirmed). -0.7 points vs the best framework (not confirmed).
459	11. Pooled summaries	multiple-choice accuracy	Best framework + Head A	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-4B	Head A	Video-MMMU (both tracks)	all lecture test questions	test	404	404	multiple-choice accuracy (answer letter correct)	55.7	225 / 404	3.7	+0.0 to +7.7	no	1.5	-1.5 to +4.5	no	Best framework + Head A answered 55.7% of 404 Video-MMMU (both tracks) test questions correctly (4B answer model). +3.7 points vs the starting framework (not confirmed). +1.5 points vs the best framework (not confirmed).
460	11. Pooled summaries	multiple-choice accuracy	Best framework + Head A + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head A	Video-MMMU (both tracks)	all lecture test questions	test	404	404	multiple-choice accuracy (answer letter correct)	40.6	164 / 404	2.0	-2.0 to +6.2	no	0.2	-2.7 to +3.2	no	Best framework + Head A + speech text answered 40.6% of 404 Video-MMMU (both tracks) test questions correctly (2B answer model). +2.0 points vs the starting framework (not confirmed). +0.2 points vs the best framework (not confirmed).

461	11. Pooled summaries	multiple-choice accuracy	Best framework + Head A + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head A	Video-MMMU (both tracks)	all lecture test questions	test	404	404	multiple-choice accuracy (answer letter correct)	56.9	230 / 404	5.0	+0.7 to +9.2	yes (gain)	0.5	-2.2 to +3.2	no	Best framework + Head A + speech text answered 56.9% of 404 Video-MMMU (both tracks) test questions correctly (4B answer model). +5.0 points vs the starting framework (confirmed). +0.5 points vs the best framework (not confirmed).
462	11. Pooled summaries	multiple-choice accuracy	Standard rule: 4 evenly spaced pictures	none	4 evenly spaced pictures	top pictures by match to the question	4	none	Qwen3-VL-2B	none	Video-MMMU (both tracks)	all lecture test questions	test	404	404	multiple-choice accuracy (answer letter correct)	36.9	149 / 404	-1.7	-6.4 to +3.0	no	-3.2	-7.4 to +1.0	no	Standard rule: 4 evenly spaced pictures answered 36.9% of 404 Video-MMMU (both tracks) test questions correctly (2B answer model). -1.7 points vs the starting framework (not confirmed). -3.2 points vs the best framework (not confirmed).
463	11. Pooled summaries	multiple-choice accuracy	Starting framework (v2 design)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	Video-MMMU (both tracks)	all lecture test questions	test	404	404	multiple-choice accuracy (answer letter correct)	38.6	156 / 404				-1.5	-5.4 to +2.5	no	Starting framework (v2 design) answered 38.6% of 404 Video-MMMU (both tracks) test questions correctly (2B answer model). -1.5 points vs the best framework (not confirmed).
464	11. Pooled summaries	multiple-choice accuracy	Starting framework (v2 design)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-4B	none	Video-MMMU (both tracks)	all lecture test questions	test	404	404	multiple-choice accuracy (answer letter correct)	52.0	210 / 404				-2.2	-6.4 to +2.0	no	Starting framework (v2 design) answered 52.0% of 404 Video-MMMU (both tracks) test questions correctly (4B answer model). -2.2 points vs the best framework (not confirmed).
465	11. Pooled summaries	multiple-choice accuracy	Best framework + confidence choice (4 sets, most confident answer)	speech rule and Head A (both pools)	hybrid	confidence choice over 4 picture sets (most confident answer)	4 (per set)	none	Qwen3-VL-2B	Head A	Video-MMMU (both tracks)	all lecture test questions	test	404	404	multiple-choice accuracy (answer letter correct)	41.6	168 / 404	3.0	-0.7 to +6.7	no	1.5	-1.2 to +4.5	no	Best framework + confidence choice (4 sets, most confident answer) answered 41.6% of 404 Video-MMMU (both tracks) test questions correctly (2B answer model). +3.0 points vs the starting framework (not confirmed). +1.5 points vs the best framework (not confirmed).
466	11. Pooled summaries	multiple-choice accuracy	Best framework + confidence choice (4 sets, summed probabilities)	speech rule and Head A (both pools)	hybrid	confidence choice over 4 picture sets (sum of option probabilities)	4 (per set)	none	Qwen3-VL-2B	Head A	Video-MMMU (both tracks)	all lecture test questions	test	404	404	multiple-choice accuracy (answer letter correct)	41.3	167 / 404	2.7	-1.0 to +6.4	no	1.2	-1.5 to +3.7	no	Best framework + confidence choice (4 sets, summed probabilities) answered 41.3% of 404 Video-MMMU (both tracks) test questions correctly (2B answer model). +2.7 points vs the starting framework (not confirmed). +1.2 points vs the best framework (not confirmed).
467	11. Pooled summaries	multiple-choice accuracy	Standard rule: dense MobileCLIP top-4	none	dense whole-video search, top 4	top pictures by match to the question	4	none	Qwen3-VL-2B	none	CG-Bench + Video-MMMU (both tracks)	all test questions	test	459	410	multiple-choice accuracy (answer letter correct)	36.8	169 / 459	-1.3	-5.1 to +2.6	no	-4.6	-7.9 to -1.3	yes (loss)	Standard rule: dense MobileCLIP top-4 answered 36.8% of 459 CG-Bench + Video-MMMU (both tracks) test questions correctly (2B answer model). -1.3 points vs the starting framework (not confirmed). -4.6 points vs the best framework (confirmed).
468	11. Pooled summaries	multiple-choice accuracy	Best framework + Head B	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-2B	Head B	CG-Bench + Video-MMMU (both tracks)	all test questions	test	459	410	multiple-choice accuracy (answer letter correct)	41.4	190 / 459	3.3	-0.2 to +6.7	no	0.0	-3.0 to +3.1	no	Best framework + Head B answered 41.4% of 459 CG-Bench + Video-MMMU (both tracks) test questions correctly (2B answer model). +3.3 points vs the starting framework (not confirmed). +0.0 points vs the best framework (not confirmed).
469	11. Pooled summaries	multiple-choice accuracy	Best framework + Head B	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-4B	Head B	CG-Bench + Video-MMMU (both tracks)	all test questions	test	459	410	multiple-choice accuracy (answer letter correct)	54.2	249 / 459	3.9	+0.4 to +7.6	yes (gain)	1.5	-2.5 to +5.4	no	Best framework + Head B answered 54.2% of 459 CG-Bench + Video-MMMU (both tracks) test questions correctly (4B answer model). +3.9 points vs the starting framework (confirmed). +1.5 points vs the best framework (not confirmed).
470	11. Pooled summaries	multiple-choice accuracy	Best framework + Head B + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head B	CG-Bench + Video-MMMU (both tracks)	all test questions	test	459	410	multiple-choice accuracy (answer letter correct)	41.0	188 / 459	2.8	-0.9 to +6.6	no	-0.4	-3.6 to +2.8	no	Best framework + Head B + speech text answered 41.0% of 459 CG-Bench + Video-MMMU (both tracks) test questions correctly (2B answer model). +2.8 points vs the starting framework (not confirmed). -0.4 points vs the best framework (not confirmed).
471	11. Pooled summaries	multiple-choice accuracy	Best framework + Head B + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head B	CG-Bench + Video-MMMU (both tracks)	all test questions	test	459	410	multiple-choice accuracy (answer letter correct)	53.8	247 / 459	3.5	-0.4 to +7.4	no	-1.7	-5.4 to +1.9	no	Best framework + Head B + speech text answered 53.8% of 459 CG-Bench + Video-MMMU (both tracks) test questions correctly (4B answer model). +3.5 points vs the starting framework (not confirmed). -1.7 points vs the best framework (not confirmed).

472	11. Pooled summaries	multiple-choice accuracy	Best framework	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	none	CG-Bench + Video-MMMU (both tracks)	all test questions	test	459	410	multiple-choice accuracy (answer letter correct)	41.4	190 / 459	3.3	-0.7 to +7.1	no				Best framework answered 41.4% of 459 CG-Bench + Video-MMMU (both tracks) test questions correctly (2B answer model). +3.3 points vs the starting framework (not confirmed).
473	11. Pooled summaries	multiple-choice accuracy	Best framework	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-4B	none	CG-Bench + Video-MMMU (both tracks)	all test questions	test	459	410	multiple-choice accuracy (answer letter correct)	52.7	242 / 459	2.4	-1.5 to +6.4	no				Best framework answered 52.7% of 459 CG-Bench + Video-MMMU (both tracks) test questions correctly (4B answer model). +2.4 points vs the starting framework (not confirmed).
474	11. Pooled summaries	multiple-choice accuracy	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	none	CG-Bench + Video-MMMU (both tracks)	all test questions	test	459	410	multiple-choice accuracy (answer letter correct)	41.4	190 / 459	3.3	-0.7 to +7.2	no				Best framework + speech text answered 41.4% of 459 CG-Bench + Video-MMMU (both tracks) test questions correctly (2B answer model). +3.3 points vs the starting framework (not confirmed).
475	11. Pooled summaries	multiple-choice accuracy	Best framework + speech text	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	none	CG-Bench + Video-MMMU (both tracks)	all test questions	test	459	410	multiple-choice accuracy (answer letter correct)	55.6	255 / 459	5.2	+1.1 to +9.5	yes (gain)				Best framework + speech text answered 55.6% of 459 CG-Bench + Video-MMMU (both tracks) test questions correctly (4B answer model). +5.2 points vs the starting framework (confirmed).
476	11. Pooled summaries	multiple-choice accuracy	No pictures (question and options only)	speech rule: MiniLM on lines inside the 4 BM25 windows	hybrid: 3 inside windows + 3 dense whole video	no pictures	0	none	Qwen3-VL-2B	none	CG-Bench + Video-MMMU (both tracks)	all test questions	test	459	410	multiple-choice accuracy (answer letter correct)	22.2	102 / 459	-15.9	-20.4 to -11.4	yes (loss)	-19.2	-23.8 to -14.4	yes (loss)	No pictures (question and options only) answered 22.2% of 459 CG-Bench + Video-MMMU (both tracks) test questions correctly (2B answer model). -15.9 points vs the starting framework (confirmed). -19.2 points vs the best framework (confirmed).
477	11. Pooled summaries	multiple-choice accuracy	Best framework + Head A + Head B	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-2B	Head A, Head B	CG-Bench + Video-MMMU (both tracks)	all test questions	test	459	410	multiple-choice accuracy (answer letter correct)	41.4	190 / 459	3.3	-0.2 to +6.8	no	0.0	-3.1 to +3.3	no	Best framework + Head A + Head B answered 41.4% of 459 CG-Bench + Video-MMMU (both tracks) test questions correctly (2B answer model). +3.3 points vs the starting framework (not confirmed). +0.0 points vs the best framework (not confirmed).
478	11. Pooled summaries	multiple-choice accuracy	Best framework + Head A + Head B	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	none	Qwen3-VL-4B	Head A, Head B	CG-Bench + Video-MMMU (both tracks)	all test questions	test	459	410	multiple-choice accuracy (answer letter correct)	54.9	252 / 459	4.6	+0.9 to +8.4	yes (gain)	2.2	-1.3 to +5.8	no	Best framework + Head A + Head B answered 54.9% of 459 CG-Bench + Video-MMMU (both tracks) test questions correctly (4B answer model). +4.6 points vs the starting framework (confirmed). +2.2 points vs the best framework (not confirmed).
479	11. Pooled summaries	multiple-choice accuracy	Best framework + Head A + Head B + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head A, Head B	CG-Bench + Video-MMMU (both tracks)	all test questions	test	459	410	multiple-choice accuracy (answer letter correct)	41.0	188 / 459	2.8	-0.8 to +6.4	no	-0.4	-3.7 to +3.0	no	Best framework + Head A + Head B + speech text answered 41.0% of 459 CG-Bench + Video-MMMU (both tracks) test questions correctly (2B answer model). +2.8 points vs the starting framework (not confirmed). -0.4 points vs the best framework (not confirmed).
480	11. Pooled summaries	multiple-choice accuracy	Best framework + Head A + Head B + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	Head B (trained, chosen version)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head A, Head B	CG-Bench + Video-MMMU (both tracks)	all test questions	test	459	410	multiple-choice accuracy (answer letter correct)	54.9	252 / 459	4.6	+0.9 to +8.3	yes (gain)	-0.7	-4.1 to +2.8	no	Best framework + Head A + Head B + speech text answered 54.9% of 459 CG-Bench + Video-MMMU (both tracks) test questions correctly (4B answer model). +4.6 points vs the starting framework (confirmed). -0.7 points vs the best framework (not confirmed).
481	11. Pooled summaries	multiple-choice accuracy	Best framework + Head A	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-2B	Head A	CG-Bench + Video-MMMU (both tracks)	all test questions	test	459	410	multiple-choice accuracy (answer letter correct)	39.7	182 / 459	1.5	-2.5 to +5.4	no	-1.7	-4.5 to +1.1	no	Best framework + Head A answered 39.7% of 459 CG-Bench + Video-MMMU (both tracks) test questions correctly (2B answer model). +1.5 points vs the starting framework (not confirmed). -1.7 points vs the best framework (not confirmed).

482	11. Pooled summaries	multiple-choice accuracy	Best framework + Head A	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	none	Qwen3-VL-4B	Head A	CG-Bench + Video-MMMU (both tracks)	all test questions	test	459	410	multiple-choice accuracy (answer letter correct)	54.0	248 / 459	3.7	-0.2 to +7.6	no	1.3	-1.8 to +4.4	no	Best framework + Head A answered 54.0% of 459 CG-Bench + Video-MMMU (both tracks) test questions correctly (4B answer model). +3.7 points vs the starting framework (not confirmed). +1.3 points vs the best framework (not confirmed).
483	11. Pooled summaries	multiple-choice accuracy	Best framework + Head A + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-2B	Head A	CG-Bench + Video-MMMU (both tracks)	all test questions	test	459	410	multiple-choice accuracy (answer letter correct)	40.5	186 / 459	2.4	-1.5 to +6.3	no	-0.9	-3.7 to +2.0	no	Best framework + Head A + speech text answered 40.5% of 459 CG-Bench + Video-MMMU (both tracks) test questions correctly (2B answer model). +2.4 points vs the starting framework (not confirmed). -0.9 points vs the best framework (not confirmed).
484	11. Pooled summaries	multiple-choice accuracy	Best framework + Head A + speech text	Head A (trained) over the whole transcript	hybrid: 3 inside windows + 3 dense whole video	MMR rule on question + options (best framework's picker)	4	speech around the chosen pictures + best lines (word + meaning search), 600 words	Qwen3-VL-4B	Head A	CG-Bench + Video-MMMU (both tracks)	all test questions	test	459	410	multiple-choice accuracy (answer letter correct)	55.8	256 / 459	5.4	+1.6 to +9.3	yes (gain)	0.2	-2.6 to +3.1	no	Best framework + Head A + speech text answered 55.8% of 459 CG-Bench + Video-MMMU (both tracks) test questions correctly (4B answer model). +5.4 points vs the starting framework (confirmed). +0.2 points vs the best framework (not confirmed).
485	11. Pooled summaries	multiple-choice accuracy	Standard rule: 4 evenly spaced pictures	none	4 evenly spaced pictures	top pictures by match to the question	4	none	Qwen3-VL-2B	none	CG-Bench + Video-MMMU (both tracks)	all test questions	test	459	410	multiple-choice accuracy (answer letter correct)	36.4	167 / 459	-1.7	-6.2 to +2.6	no	-5.0	-8.8 to -1.1	yes (loss)	Standard rule: 4 evenly spaced pictures answered 36.4% of 459 CG-Bench + Video-MMMU (both tracks) test questions correctly (2B answer model). -1.7 points vs the starting framework (not confirmed). -5.0 points vs the best framework (confirmed).
486	11. Pooled summaries	multiple-choice accuracy	Starting framework (v2 design)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-2B	none	CG-Bench + Video-MMMU (both tracks)	all test questions	test	459	410	multiple-choice accuracy (answer letter correct)	38.1	175 / 459				-3.3	-7.1 to +0.7	no	Starting framework (v2 design) answered 38.1% of 459 CG-Bench + Video-MMMU (both tracks) test questions correctly (2B answer model). -3.3 points vs the best framework (not confirmed).
487	11. Pooled summaries	multiple-choice accuracy	Starting framework (v2 design)	speech rule: MiniLM on lines inside the 4 BM25 windows	scan every 5 s inside the BM25 windows	top pictures by match to the question	4	BM25 excerpt, 120 words	Qwen3-VL-4B	none	CG-Bench + Video-MMMU (both tracks)	all test questions	test	459	410	multiple-choice accuracy (answer letter correct)	50.3	231 / 459				-2.4	-6.4 to +1.5	no	Starting framework (v2 design) answered 50.3% of 459 CG-Bench + Video-MMMU (both tracks) test questions correctly (4B answer model). -2.4 points vs the best framework (not confirmed).
488	11. Pooled summaries	multiple-choice accuracy	Best framework + confidence choice (4 sets, most confident answer)	speech rule and Head A (both pools)	hybrid	confidence choice over 4 picture sets (most confident answer)	4 (per set)	none	Qwen3-VL-2B	Head A	CG-Bench + Video-MMMU (both tracks)	all test questions	test	459	410	multiple-choice accuracy (answer letter correct)	42.3	194 / 459	4.1	+0.5 to +7.9	yes (gain)	0.9	-1.6 to +3.3	no	Best framework + confidence choice (4 sets, most confident answer) answered 42.3% of 459 CG-Bench + Video-MMMU (both tracks) test questions correctly (2B answer model). +4.1 points vs the starting framework (confirmed). +0.9 points vs the best framework (not confirmed).
489	11. Pooled summaries	multiple-choice accuracy	Best framework + confidence choice (4 sets, summed probabilities)	speech rule and Head A (both pools)	hybrid	confidence choice over 4 picture sets (sum of option probabilities)	4 (per set)	none	Qwen3-VL-2B	Head A	CG-Bench + Video-MMMU (both tracks)	all test questions	test	459	410	multiple-choice accuracy (answer letter correct)	42.9	197 / 459	4.8	+0.9 to +8.5	yes (gain)	1.5	-0.9 to +3.9	no	Best framework + confidence choice (4 sets, summed probabilities) answered 42.9% of 459 CG-Bench + Video-MMMU (both tracks) test questions correctly (2B answer model). +4.8 points vs the starting framework (confirmed). +1.5 points vs the best framework (not confirmed).